

Temporal Dynamics of the Plasma Proteomic Landscape Reveals Maladaptation in ME/CFS Following Exertion

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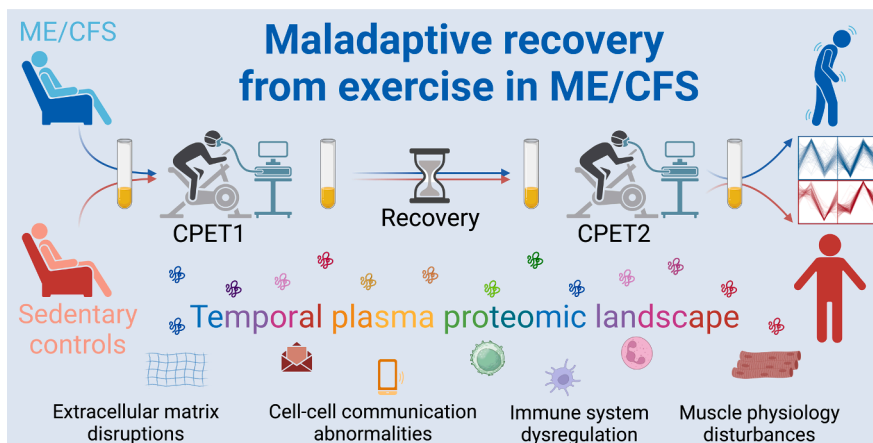
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Graphical Abstract

In Brief

To investigate the molecular basis of post-exertional malaise in ME/CFS, we performed longitudinal plasma proteomics in 132 individuals undergoing two maximal exercise tests. ME/CFS patients showed persistent immune, metabolic, and neuromuscular dysregulation during recovery, including suppression of T and B cell signaling and activation of glycolytic stress pathways. Protein changes were linked to symptom severity and reduced exercise capacity, with distinct sex-specific responses. These findings highlight impaired molecular recovery from exertion in ME/CFS and support the use of time-resolved proteomics to characterize post-exertional pathology.



Highlights

- Plasma profiling of 7288 proteins during post-exertional malaise in ME/CFS.
- ME/CFS participants show sustained immune, metabolic, and neuromuscular dysregulation during post-exercise recovery.
- Exertion disrupts T and B cell signaling, IL-17 pathways, and mitochondrial metabolism.
- Protein signatures correlate with symptom severity and impaired exercise performance in patients with ME/CFS.
- Sex-stratified analysis reveals distinct molecular responses, underscoring the importance of sex in ME/CFS pathophysiology.

Temporal Dynamics of the Plasma Proteomic Landscape Reveals Maladaptation in ME/CFS Following Exertion

Arnaud Germain¹, Katherine A. Glass¹, Melissa A. Eckert¹, Ludovic Giloteaux¹, and Maureen R. Hanson^{1*}

The overarching symptom of myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is post-exertional malaise (PEM), an exacerbation of symptoms following physical or mental exertion. To investigate the molecular underpinnings of PEM, we performed longitudinal plasma proteomics using the Somascan 7K aptamer-based assay to monitor 6361 unique plasma proteins in 132 individuals (96 females and 36 males) subjected to two maximal cardiopulmonary exercise tests separated by a 24-h recovery period. The cohort included 79 ME/CFS cases compared to 53 age- and BMI-matched sedentary controls, allowing us to distinguish disease-specific molecular alterations from those due to physical deconditioning. Longitudinal profiling revealed widespread proteomic changes following exertion, with the most pronounced alterations observed in ME/CFS participants during the recovery phase, coinciding with the onset of PEM. Compared to controls, patients with ME/CFS showed persistent dysregulation of immune, metabolic, and neuromuscular pathways. Key findings included suppression of T and B cell signaling, downregulation of IL-17 and cell-cell communication pathways, and upregulation of glycolysis/gluconeogenesis, suggestive of mitochondrial stress and impaired immune recovery from exercise. Proteomic associations with physiological performance (VO_2max , anaerobic threshold) revealed disruptions between protein abundance and exercise capacity in ME/CFS versus controls. Correlations with symptom severity linked changes in immune-related proteins and ME/CFS symptoms, including muscle pain, recurrent sore throat, and lymph node tenderness. Sex-stratified analyses revealed distinct molecular responses between females and males, emphasizing the importance of considering sex as a biological variable in ME/CFS research. Finally, our analysis of sedentary controls contributes new data on molecular responses to acute exertion in a predominantly female sedentary cohort, a population historically underrepresented in exercise physiology studies. Together, these findings underscore the value of dynamic, proteomic

profiling over time for characterizing maladaptive responses to exertion in ME/CFS and provide a foundation for deeper mechanistic investigation into PEM.

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a disabling chronic illness that inflicts a slew of symptoms, including but not limited to fatigue, cognitive problems, and muscle and joint pain, that are not alleviated by rest or sleep (1). Symptom severity constrains the sufferers' lives, preventing them from engaging fully in daily life, with disability levels ranging from 'no symptoms at rest' to 'severe symptoms on a continuous basis', sometimes leaving individuals bedridden, hypersensitive to external stimuli, and incapable of self-care (2). Most patients with ME/CFS report that post-exertional malaise (PEM) is the most disabling symptom, as it prevents "pushing through" the other symptoms. PEM is the exacerbation of various symptoms when personal physical, mental and/or emotional exertion thresholds are exceeded (3). While exercise is generally promoted for maintaining physical health and managing many conditions (4–6), this approach can be counterproductive for ME/CFS patients, in which the body's response to exercise exacerbates symptoms. Instead, careful pacing is crucial to avoid triggering PEM (7), as repeated episodes can lead to a permanent worsening of the illness and loss of physical and mental functional capacity.

To investigate PEM at the molecular level, we generated a high-throughput plasma protein profiling dataset with four blood draws over a 2-day cardiopulmonary exercise test (CPET) protocol. This protocol was designed to create a significant exercise challenge to induce PEM in volunteer patients with ME/CFS (8). The plasma samples used in this study were also analyzed in our large-scale metabolomics study (9), with none of the samples collected post-COVID-19 infection. Our goal was to scrutinize the plasma proteome

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response to exercise in patients with ME/CFS compared to sedentary controls.

Using the high-plex, high-throughput 7K Somascan assay (SomaLogic) (10, 11), our proteomics dataset spans a 10-log dynamic range from femtomolar [10^{-15}] to micromolar [10^{-6}] concentrations, resulting in over 3.6 million total datapoints for 6361 unique proteins recognized by 7288 aptamers (SOMAmers). The proteins analyzed are distributed proportionately with the human proteome in terms of cellular compartment, and provide robust coverage of key biological pathways, including metabolism, genetic information processing, environmental information processing, cellular processes, organismal systems, and diseases. Assay reliability has been carefully documented by SomaLogic (12), as well as by many other groups using this platform (13, 14). This dataset represents an increase of 1571 proteins compared to our previous pilot proteomics study in 20 female patients with ME/CFS and 20 controls (15), which was the first ME/CFS proteomics dataset generated using SOMAmer technology from SomaLogic. An NIH initiative recently released a smaller dataset (1305 protein analytes) but found no differences between the small cohorts of 21 controls and the 17 ME/CFS subjects enrolled (16). Several characteristics set our dataset apart in the ME/CFS proteomics field, including a larger cohort size (132 participants), the inclusion of both females and males (96 and 36, respectively), and, most importantly, the recruitment of sedentary controls as well as the longitudinal sampling spanning two exercise challenges. A recent publication of exercise physiology data from a larger segment of the same 173-participant cohort (NCT04026425–ClinicalTrials.gov), underscores the impact of PEM on the exercise recovery of individuals with ME/CFS (17). Results from the specific symptom severity (SSS) score survey, collected before the CPETs and monitored over the subsequent 10 days, confirm the substantial impact of the illness on the ME/CFS cohort selected for this proteomics study.

The analysis of our extensive dataset reveals no significant differences in individual protein levels between patients with ME/CFS and sedentary controls at baseline, and only a few differences at specific timepoints following our acute exercise protocol. However, quantitative pathway analysis for these same timepoints identified disruptions in immune system pathways, cell-cell communication, and energy metabolism. Importantly, the temporal dynamics of our study reveal significant maladaptation in patients with ME/CFS compared to controls, particularly in proteins involved in the immune system, signal transduction, and muscle contraction. Notably, some of the most striking differences emerged following analysis of male and female cohorts separately. We concluded our analysis by identifying many associations between plasma protein levels and clinical and exercise physiology parameters.

EXPERIMENTAL PROCEDURES

Cohort Selection

Our cohort of 132 patients is an expansion by 25 patients of the cohort profiled in our previous large-scale plasma metabolomics study (9). Both are part of the larger 173-participant cohort NCT04026425 (ClinicalTrials.gov), where all ME/CFS patients met the Canadian consensus criteria for ME/CFS diagnosis (18), and the sedentary controls were recruited by the same physicians at the same three sites: Ithaca College, Ithaca, NY; Weill Cornell Medicine, NYC, NY; and Workwell Foundation, Torrance, CA. All samples were collected prior to the COVID-19 pandemic shutdown in March 2020. The cohort features are summarized in Figure 1B and more extensively described in Supplemental Table S1.

Maximal-Effort Exercise Test Protocol

As illustrated in Figure 1A and reported previously (8), all patients completed a maximal-effort CPET, pedaling on a stationary cycle until volitional exhaustion, as indicated by a respiratory exchange ratio (RER) > 1.1. Following 24 h of rest, the same maximal-effort CPET was repeated with the intent to document PEM induction in participants with ME/CFS.

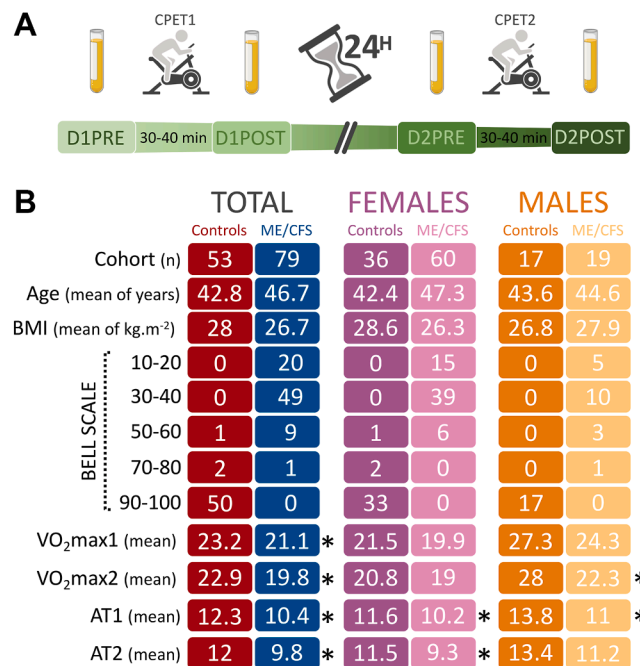


FIG. 1. Study design and population statistics. A, overview of the blood collection timeline encompassing two maximal cardiopulmonary exercise tests (CPETs), each separated by a 24-h recovery period. B, summary of the most representative features of the population. A score of 100 on the Bell scale indicates no symptoms at rest or during exercise, normal activity level and the ability to work full-time without difficulties. Conversely, a score of 0 signifies continuous symptoms, being bedbound, and inability to self-care. CPET peak VO₂ (VO₂max1/2) and CPET anaerobic threshold (AT1/2) were assessed on Day1 and Day2, respectively, with values expressed in ml.kg⁻¹.min⁻¹. An asterisk (*) denotes statistical significance at $p < 0.05$ based on a t test.

Blood Collection

All plasma samples were isolated as previously described by our group (9), from venous blood collected in EDTA tubes at four longitudinal timepoints, encircling the two maximal-effort CPETs interrupted by a 24-h recovery period (Fig. 1A). Most samples were collected in the morning, and participants were asked to stop any supplement medications 2 weeks before the CPETs, and to stop pain medications and stimulants 2 days before.

Proteomics Dataset

Large-scale protein profiling was performed by SomaLogic, Inc. in Boulder, CO (www.somallogic.com) as previously done for an exploratory study published by our group (15). At the time the plasma samples were shipped, the version of their technology named SomaScan Assay relied on 7288 SOMAmers (Slow Offrate Modified Aptamers), designed to target 6361 unique human proteins. A dynamic range from femtomolar to micromolar is achieved through three dilutions, with 6001 proteins profiled at 20% dilution, 1100 at 0.5% dilution, and 187 at 0.005% dilution. The generated dataset did not have any missing data. One sample, COR-4806 D1PRE, was excluded due to a technical issue during the SomaLogic assay.

Data Processing

The SomaScan Assay data standardization performed by SomaLogic accounts for various aspects of the assay as detailed hereafter. The first normalization uses hybridization controls to mitigate variation within the run that comes from the readout steps: transfer to Agilent slides, hybridization, wash, and scan. It is then followed by median signal normalization across pooled calibrator replicates within the run to mitigate within-run technical variation in the calibrator signal prior to use in scaling calculations. The set of ratios of the calibrator reference value to the median of calibrator replicates for each SOMAmer reagent is calculated and decomposed into two terms: plate scale - the median ratio, and calibration scale - the recalculated set of scale factors, one for each SOMAmer reagent. Plate scale adjusts for overall signal intensity differences between runs. Calibration adjusts for the SOMAmer reagent-specific assay differences between runs. Median signal normalization is performed using Adaptive Normalization by Maximum Likelihood (ANML) when the specimen type and study population align with pre-defined SomaLogic reference distributions. When this alignment is not met, a more conservative median normalization to a study-specific reference group is applied instead. As recommended by SomaLogic, we used the ANML normalization for all analyses.

Additionally, based on SomaLogic's acceptance criteria, 477 SOMAmers (6.5%) were "flagged" for quality control concerns. Of these, 409 (86%) were from the 20% dilution, 45 (9%) from the 0.5% dilution, and 23 (5%) from the 0.005% dilution. The flagging process, referred to as 'ColCheck', is a quality control tool to monitor assay performance by identifying SOMAmer reagents with scale factors outside the acceptable range (0.8–1.2) in their positive control replicates. As detailed by a SomaLogic in a technical note (https://dam.somallogic.com/m/34de2ede028f756b/original/D00006601_Rev-1_2_023-12_Flagged-Sample-Tech-Note.pdf), these reagents are labeled as 'Pass' or 'Flag' based solely on quality control metrics and do not impact sample data integrity or analyte inclusion in the study. Flagged SOMAmers are typically the result of excessive scaling factors for normalization or calibration but are not removed from analysis unless later evidence warrants elimination. Therefore, all measurements were retained in our analysis.

Data Analysis and Statistics

Univariate statistical analysis for each SOMAmer was performed using either a linear model or linear mixed model with fixed effects of disease status, age, BMI, site and sex, and a random effect of subject. The Benjamini-Hochberg (BH) method was used to correct for false discovery rate at a threshold of $q < 0.05$ (FDR) (19).

GSEA

The GSEA analysis was performed in R using the *multiGSEA* package. This package is a wrapper for the *fgsea* package. The input ranked list of all unique proteins (using UniProt IDs) was prepared by getting the median for each phenotype at each time point followed by the calculation of the log2 Fold Change (log2FC) for ME/CFS vs controls. For proteins targeted by multiple SOMAmers, the SOMAmer with the highest |log2FC| was used in the analysis to avoid duplicates. The 'fgsea' function parameters were as follows: $\text{eps} = 0$, $\text{minSize} = 15$, $\text{maxSize} = 500$, $\text{nPermSimple} = 50,000$. Significant pathways had q -values < 0.05 obtained following BH FDR correction.

Fuzzy c-means Clustering and Elbow Method

The clustering was performed in R (20) using the *Mfuzz* package. Protein selection was based on the linear mixed model q -value results with a cutoff at 0.05. The proteomics input data was prepared by calculating the median for each phenotype at each time point, followed by a scaling of the data to make the protein data comparable using the 'standardize' function from the *reshape2* library. The number of clusters was determined using the elbow method by calculating the within sum of squared errors (WSS) for cluster results and plotting the WSS values for different numbers of clusters.

Pathway Enrichment Analysis of Clusters

Proteins significantly different within each of the fuzzy c-means clusters were analyzed for pathway enrichment analysis using Enrichr with the Reactome database (<https://maayanlab.cloud/Enrichr/>) (21–23). Statistical significance is assessed with Fisher's exact test under the assumption of gene independence, and the q -value (adjusted p -value) is obtained using the Benjamini-Hochberg method.

t-SNE Clustering and Cut-Off Two-Component Model

The t-SNE plot was generated in R using the *Rtsne* package and visualized with *ggplot*, with each point colored by participant and shaped by timepoint. Pairwise Euclidian distances between each participant's time point were calculated in the two-dimensional t-SNE space. A two-component Gaussian mixture model was then fit to the distribution of these distances, using the *mixtools* R package. The cutoff distance, defined as the point where the probability of membership in each component is equal, was estimated at 1.7455 (see Supplemental Fig. S1).

Logistic Regression With Protein t-SNE Clusters

Logistic regression analysis was performed to explore factors associated with the SomaLogic t-SNE binary cluster assignments (for ΔDay1 , Recovery, ΔDay2) using the following parameters: age, BMI, SF-36 general health (GH), SF-36 mental component summary (MCS), disease status, sex, test site, CPET day 1 peak VO_2 ($\text{VO}_{2\text{max}1}$), CPET day 2 peak VO_2 ($\text{VO}_{2\text{max}2}$), CPET day 1 anaerobic threshold (AT1), and CPET day 2 anaerobic threshold (AT2) (*stats* R package). Due to collinearity, the variables SF-36 physical component summary (PCS) and weight were removed (Pearson correlations with $R > 0.5$), and the exercise physiology variables were tested in the model one at a time. The response variable for the regression model

was the t-SNE cluster assignment with values 0 or 1, based on the similarity between t-SNE components 1 and 2 for each time point with 0 indicating high intra-subject similarity and 1 indicating low intra-subject similarity. *p*-values were obtained by the Wald test (also known as Z-test) to determine the significance of predictors in the model with the null hypothesis that there is no association between cluster assignment and the predictor (24). Odds ratios were calculated by exponentiating the coefficient associated with each predictor.

Correlation of Protein Levels With Clinical Parameters in Patients With ME/CFS

Spearman correlation coefficients were calculated for protein levels at each timepoint versus the following parameters: ME/CFS duration (ME/CFS group only), age, BMI, SF-36 PCS, MCS, GH, Bell Activity Scale score, the percentage of waking time spent in a reclined position, the multi-dimensional fatigue inventory (MFI)-20 total score (in patients with ME/CFS only), and symptom severity scores. Symptom severity was assessed using specific symptom severity (SSS) surveys, in which all patients rated their symptoms on a scale of 0 to 10, (0 = not present and 10 = very severe) on average over the past month (recent), on the morning of the first CPET (D1PRE), and on the morning of the second CPET (D2PRE). The “delta” SSS scores represent the change in symptom severity from pre-exercise, calculated as follows: the scores at D2PRE – the scores at D1PRE (Delta1), the scores at 2 days post CPET 1 – D1PRE (Delta2), and the scores at 4 days post CPET 1 – D1PRE (Delta3). The following symptoms were included in the analysis: fatigue, impaired memory & concentration, recurrent sore throat, lymph node tenderness, muscle tenderness & pain (myalgia), joint pain (arthralgia), headache, unrefreshing sleep, and post-exertional malaise (PEM).

The significance of the correlation coefficients was determined via *t* test (*hmisc* R package) with the null hypothesis that the correlation coefficient *R* is equal to 0. *p*-values were adjusted using the BH FDR correction procedure. We considered significant and interesting correlations to have Spearman $|R| > 0.5$ and $q < 0.1$.

Correlation analysis with the same methods was conducted between the within-subject fold changes over time for each protein versus the same clinical and demographic parameters. The time point ratios evaluated were D1POST/D1PRE (impact of exercise on Day1), D2PRE/D1POST (recovery), D2POST/D2PRE (impact of exercise on Day2), D2PRE/D1PRE, and D2POST/D1PRE. All of these correlation analyses were done for ME/CFS and control groups separately, and repeated for total, females only and males only.

We performed Reactome pathway enrichment analysis using Enrichr software (21) for sets of proteins in which at least 60 proteins were significantly correlated with one clinical parameter at one time point or time point ratio for female patients with ME/CFS. Reactome terms and pathways were considered significantly enriched with $q < 0.05$.

Linear Regression Analysis of Protein Levels and CPET Parameters

Linear regression was used to assess the relationship between protein levels and two CPET parameters, VO_2max and anaerobic threshold (AT), both adjusted for body weight. Protein levels were standardized with a mean of 0 and a standard deviation of 1. Regression analyses were performed at each time point using the base lm function in R, with protein levels as the response variables and VO_2max and AT as predictors. Age, sex, and site were included as covariates. The *p*-values were corrected for multiple comparisons using the BH method ($q < 0.1$). The data were visually depicted with Venn diagrams using the software Venny 2.1 <http://bioinfogp.cnb.csic.es/tools/venny/index.html>.

Data Visualization

All plots were generated in R with ggplot2.

Study Approval

The study was conducted according to the guidelines of the Declaration of Helsinki. Protocols regarding human participants were approved by the Institutional Review Boards of Weill Cornell Medical College (Protocol 1708018518) and Ithaca College (IRB 1017-12Dx2). Written informed consent was received prior to participation.

RESULTS

Protocol for Inducing PEM in Participants With ME/CFS

Both patients with ME/CFS and healthy controls completed two CPETs conducted 24 h apart on a stationary cycle ergometer, designed to trigger PEM in ME/CFS participants (Fig. 1A). Each test lasted between 6 to 14 min, depending on individual effort capacity. Blood samples were collected at four timepoints: 15 to 20 min before each exercise on Day1 (D1PRE) and Day2 (D2PRE), and 15 to 20 min after each exercise on Day1 (D1POST), and Day2 (D2POST), as depicted in Figure 1A. Maximum effort during the CPET, defined by a respiratory exchange ratio (RER) > 1.1 , was reached by 96% of participants with ME/CFS on Day1 and 91% on Day2 (with no overlap in patients who did not reach the proper RER between the two CPETs), and by 98% for control participants (with one different individual at each CPET not reaching maximal effort according to RER).

The ME/CFS Cohort Demonstrates Functional and Physiological Impairments Compared to Controls

Our study combines 79 participants with ME/CFS, diagnosed according to the Canadian consensus criteria (18), and 53 sedentary but otherwise healthy controls, totaling 132 participants. Cohort demographics are summarized in Figure 1B, including sex distribution, mean age (range: 18–69 years old) and mean body mass index (BMI, range: 18.5–41.8 kg m^{-2}). Functional status was markedly lower in the ME/CFS group, as quantified using the Bell disability scale (2), which was categorized into five bins. In contrast, control participants, although sedentary, were fully functional. Our maximal exercise protocol required selecting ME/CFS participants with less symptom severity than bedbound patients. Additional clinical characteristics are detailed in Supplemental Table S1, including illness duration (mean: 11.4 years, range: 1–38 years), scores from the multidimensional fatigue inventory-20 (MFI-20), percentage of waking hours spent reclined or semi-reclined, SF-36 health survey outcomes (25), and plasma protein measurements.

Key exercise physiology data – maximal oxygen uptake (VO_2max), and anaerobic threshold (AT) from both CPET1 and CPET2, are presented in Figure 1B. While statistical significance was not consistently achieved in sex-stratified analyses, ME/CFS participants demonstrated significantly lower aerobic capacity across all exercise parameters in the overall cohort. These physiological impairments are consistent with

recent findings from a broader analysis of a larger portion of the same 173-participant cohort (NCT04026425 – [ClinicalTrials.gov](#)), which documented that ME/CFS participants failed to reproduce their CPET1 performance during CPET2, unlike matched controls. This is a marked contrast to the reproducibility typically observed in healthy, athletic, and even other diseased populations (8).

Baseline Analysis Reveals No Significantly Altered Individual Proteins in the ME/CFS Cohort

To investigate differences in protein abundance between ME/CFS patients and controls, we applied linear models at each of the four timepoints, adjusting for confounding factors (model formula: protein ~ phenotype * sex + age + BMI + site). Analyses were conducted both on the full cohort and separately for females and males (excluding sex as a variable in the sex-stratified models) to explore potential sex-specific effects. Complete results are available in [Supplemental Table S2](#) and are presented as volcano plots in [Supplemental Fig. S2](#), highlighting log₂FC of mean protein levels for patients/controls against *p*-values. Corresponding *q*-values, computed using the Benjamini-Hochberg (BH) procedure to correct for multiple comparisons, are presented in [Figure 2](#), where the significance threshold was set at *q* < 0.05.

While a number of proteins initially met the significance threshold based on uncorrected *p*-values, with a modest tendency toward decreased abundance in ME/CFS, these associations largely did not withstand adjustment for false discovery ([Fig. 2](#)). At baseline (D1PRE), post-CPET1 (D1POST), and post-CPET2 (D2POST), no proteins reached *q* < 0.05 in the full cohort, nor in either sex-stratified subgroups. However, at the D2PRE timepoint (24 h of recovery) five proteins were differentially abundant in the total cohort. Two showed significantly increased abundance in patients with ME/CFS, while three were significantly decreased. These findings suggest subtle but detectable shifts in protein expression during the recovery window.

Although no proteins reached significance in the female (*n* = 60 ME/CFS, *n* = 36 controls) and male (*n* = 18 ME/CFS, *n* = 17 controls) subgroups, the volcano plot for the male cohort at D2PRE displayed a notable enrichment of proteins just below the *q* < 0.05 threshold. This pattern may reflect sex-specific differences in recovery dynamics, although caution is warranted given the small sample size of the male subgroup.

Key Proteins Provide Novel Insights into ME/CFS Muscle and Brain Biology

The five proteins highlighted in [Figure 2](#) for the total cohort are functionally linked to muscle or brain function, offering

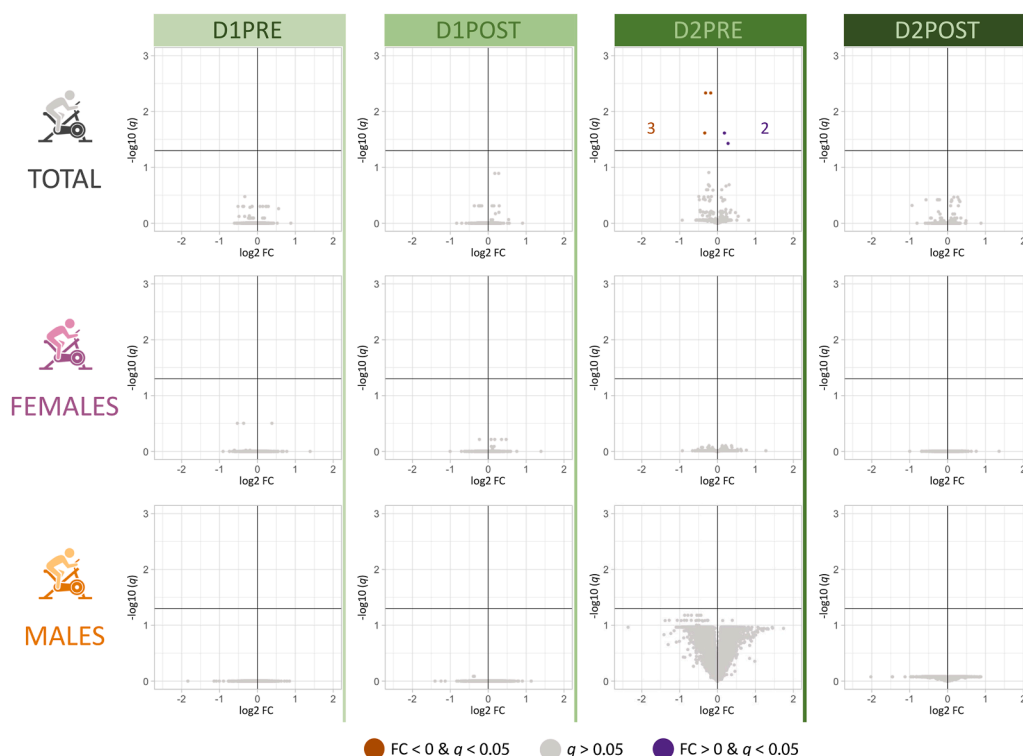


FIG. 2. Volcano plots displaying the linear modeling results comparing controls and ME/CFS patients at each timepoint (D1PRE, D1POST, D2PRE and D2POST). The analysis was performed separately for the total, female, and male cohorts. The horizontal line shows a significance threshold of *q* < 0.05, with the gray dots symbolizing non-significant proteins. Colored dots represent significantly different proteins with negative (orange) or positive (purple) log₂ fold change (FC) of the mean for patients versus controls.

potential mechanistic insights into ME/CFS pathophysiology. Their abundance patterns across timepoints are visualized in Figure 3 using box plots, stratified by total cohort, females, and male (each dot represents an individual). Notably, AHSG and CBLN4 (Fig. 3, A and B), exhibited significantly increased levels in patients with ME/CFS, while muscle-related proteins MYBPC1, KLHL41 and MYL3 (Fig. 3, C–E) showed significantly decreased levels in patients with ME/CFS at D2PRE. Our exercise protocol particularly accentuated the differences between controls and patients, peaking at D2PRE.

The male cohort showed distinct patterns, with certain box plots revealing minimal or no overlap between patients and controls (excluding whiskers and outliers) at D1PRE and D2POST for AHSG and D2PRE for MYBPC1, KLHL41, and MYL3, suggesting potential stronger or more consistent effects in males. While these findings must be interpreted with caution due to smaller sample sizes, they point to possible sex-specific differences in the proteomic response to exercise.

Secreted by the liver, AHSG (Fetuin-A or Alpha-2-HS-glycoprotein) is a multifunctional plasma agent involved in regulating insulin receptor signaling, bone remodeling, and calcium metabolism. It also inhibits pathological calcifications (26) and has been implicated in tumor progression in various cancers (27–29).

CBLN4 (Cerebellin-4) is part of a family of four proteins that form complexes with the presynaptic cell-adhesion molecules

neurexins and the postsynaptic glutamate-receptor-related proteins GluD1 and GluD2. It influences synapse connectivity and organization and promotes interneurons differentiation (30, 31). In mice, it is selectively expressed in specific neuron subsets, including excitatory cortical neurons and mitral projection neurons (31).

MYBPC1 (Myosin-binding protein C, slow type) binds to myosin and actin in skeletal muscles. Mutations are associated with severe myopathies and muscle tremors (32, 33).

KLHL41 (Kelch-like protein 41) is critical in skeletal muscle development and maintenance by regulating myoblast proliferation and differentiation (34). Mutations cause nemaline myopathy, a congenital muscle disorder (35).

MYL3 (Myosin light chain 3) is an essential component of the myosin complex in both cardiac and skeletal muscles. Declining levels have been associated with osteoarthritis (36), and its release into circulation upon muscle injury suggests it may serve as a biomarker for muscle damage across mammals (37).

Gene Set Enrichment Analysis Reveals Dysregulation in Energy Metabolism, Cell-Cell Communication, and Immune Signaling in ME/CFS

We then performed gene set enrichment analysis (GSEA), a quantitative pathway-level approach that considers the entire proteomic dataset rather than focusing only on proteins surpassing a fixed significance threshold. Proteins were ranked

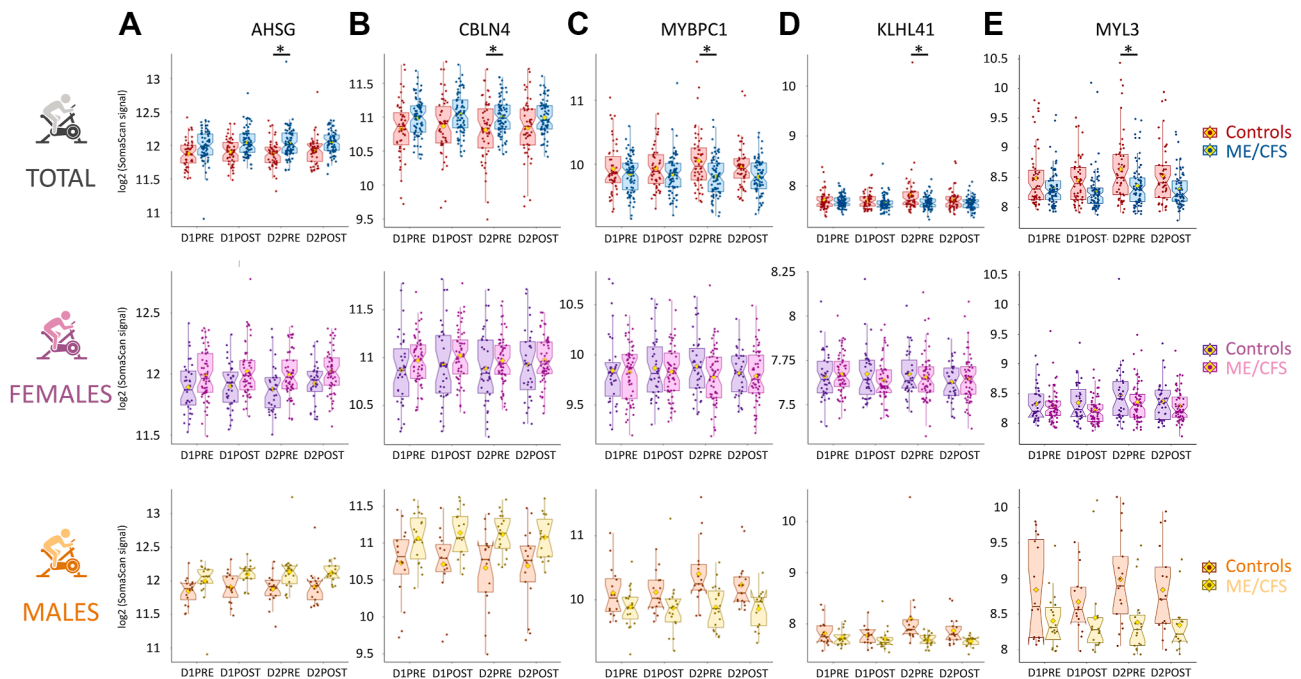


FIG. 3. Box plots of proteins significantly different between controls and patients at D1POST and D2PRE for $q < 0.15$. The labels are the EntrezGeneSymbol for each protein. Listed hereafter are the protein full names and their UniProt numbers in parenthesis. A, AHSG – Alpha-2-HS-glycoprotein (P02765). B, CBLN4 – Cerebellin-4 (Q9NTU7). C, MYBPC1 – Myosin-binding protein C, slow type (Q00872). D, KLHL41 – Kelch-like protein 41 (O60662). E, MYL3 – Myosin light chain 3 (P08590). The y-axis values are the log2-transformed values provided by SomaLogic. The yellow diamonds represent the mean. An asterisk (*) denotes statistical significance at $q < 0.05$.

by their median log₂FC (ME/CFS vs. controls) and GSEA was conducted using the KEGG database (Kyoto Encyclopedia of Genes and Genomes, www.kegg.jp) to assess functional pathway enrichment. Significantly enriched pathways were defined as those with $q < 0.05$ and an absolute normalized enrichment score (|NES|) > 1.5 (Fig. 4). Full results of the GSEA are provided in Supplemental Table S3, including pathways relating to disease and alcoholism that were considered irrelevant.

Most significantly enriched pathways had negative NES values, reflecting lower protein levels in ME/CFS relative to controls. A notable exception was the “glycolysis/gluconeogenesis” pathway at D2PRE, which had a positive NES, suggesting altered energy metabolism in ME/CFS during recovery.

At D1PRE, five pathways were significantly enriched (Fig. 4), including the “MAPK signaling”, “phospholipase D signaling”, and “sphingolipid signaling”, all central to cell-cell signaling. Other pathways were “cholinergic synapse”, involved in central nervous system (CNS) signaling, and “T cell receptor signaling”, which was the only pathway showing persistent alterations across all three timepoints.

The most extensive enrichment was observed at D2PRE, with 15 significantly altered pathways spanning neuronal signaling, metabolism, immune activation, and intracellular communication. Pathways such as “axon guidance”, “cholinergic synapse”, and “inflammatory mediator regulation of TRP channels”, highlight neuroimmune dysregulation during recovery. The enrichment of several signaling cascades (mTOR, MAPK, PI3K-Akt, Wnt, and Ras) suggest altered transduction of metabolic and inflammatory indicators. The “signaling pathways regulating pluripotency of stem cells” pathway encompasses three of these cascades (MAPK, PI3K-Akt and Wnt), reflecting shared signaling mechanisms that are not specific to stem cells. Aligning with previous ME/CFS findings, the “mTOR signaling pathway”, a central regulator of nutrient

sensing, immune activation, and cellular stress responses, was highly significantly affected, supporting a role for impaired metabolic recovery mechanisms (38–40). Disruptions in the “endocytosis” pathway related to intracellular transport, further points to widespread dysfunction in vesicle-mediated signaling. Collectively, these findings suggest D2PRE represents a key inflection point in ME/CFS recovery, marked by multisystem disruption.

By D2POST, five pathways remained significantly enriched and five additional pathways were downregulated. Four were related to immune function: “T cell receptor signaling”, “B cell receptor signaling”, and “IL-17 signaling”, and “Fc epsilon RI signaling”, the latter two being key players in both acute and chronic inflammation. Three pathways were neurologically relevant, including “axon guidance”, “dopaminergic synapse”, and “inflammatory mediator regulation of TRP channels”, further supporting ongoing neuroimmune involvement during late recovery. The remaining two pathways, (focal adhesion) and (gap junction), intersect through shared regulation of cytoskeletal dynamics and cell-cell communication.

Taken together, these results reveal that while individual protein changes may be subtle, robust systems-level dysregulation is evident, particularly in pathways governing metabolism, immune signaling, and neuronal communication. These findings are consistent with the multisystem nature of ME/CFS and underscore the value of pathway-level analyses in uncovering disease-specific biology.

Temporal Analysis Reveals Pronounced Recovery Differences Between ME/CFS and Control Cohorts

To examine longitudinal changes in protein levels in response to exercise and during recovery, we used a linear mixed-effects model, incorporating subject-level random effects to account for repeated measures (model formula: protein ~ phenotype * timepoint + disease status * sex + age + BMI + site + (1 | subject)). This approach enabled

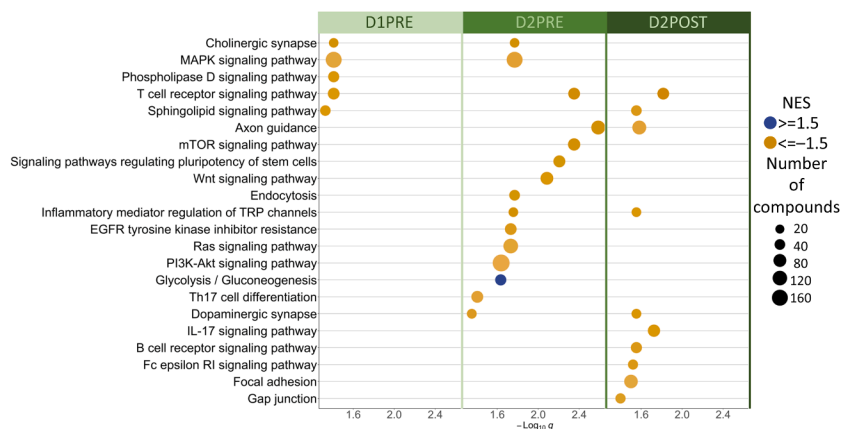


FIG. 4. GSEA analysis using the KEGG database for the total cohort – The threshold for the normalized enrichment score (NES) was set at 1.5 and the significance at $q < 0.05$. Pathways are displayed in order of most to least significant for individual time points beginning with D1PRE, with the x-axis showing significance. The bubble color shows the NES score and the bubble size shows the number of compounds.

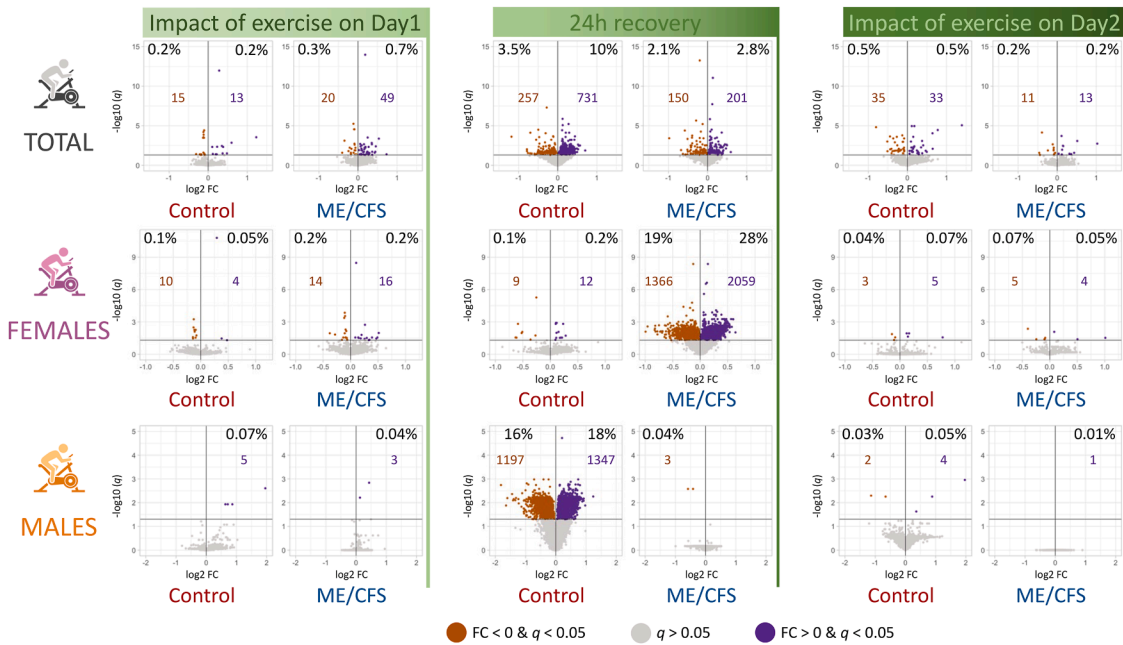


FIG. 5. Volcano plots displaying the linear mixed modeling results testing for interactions between disease status (controls and ME/CFS) and some of the timepoints (D1POST vs. D1PRE – ‘Impact of exercise on Day1’; D2PRE vs. D1POST – ‘24h recovery’; and D2POST versus D2PRE – ‘Impact of exercise on Day2’). The analysis was performed separately for the total, female, and male cohorts. The horizontal line shows a significant threshold at $q < 0.05$, with the gray dots symbolizing non-significant protein changes. Colored dots represent proteins significantly changing contingent on the negative (orange) or positive (purple) log2 fold change of each timepoint combination.

us to model within-cohort temporal protein dynamics while controlling for key covariates. **Figure 5** presents volcano plots comparing three timepoint transitions within each cohort: the impact of exercise on Day1 (D1POST vs. D1PRE), the 24-h recovery period (D2PRE vs. D1POST), and the impact of exercise on Day2 (D2POST vs. D2PRE). Additional pairwise comparisons are provided in **Supplemental Fig. S3**. All analyses were conducted on the total cohort and stratified by sex and are available in **Supplemental Table S2**.

The immediate post-exercise responses (D1POST vs. D1PRE and D2POST vs. D2PRE) affected fewer than 1% of measured proteins in any group (**Fig. 5**). This modest response likely reflects the short interval between pre- and post-exercise blood draws (30–40 min, **Fig. 1A**). A similar result was observed for the longer interval between D1PRE to D2POST (**Supplemental Fig. S3**).

In contrast, the most striking changes occurred during the 24-h recovery period (D1POST to D2PRE). At this interval, 13.5% of the proteins in the control cohort and 4.9% in the ME/CFS cohort showed significant changes, with a set of 241 proteins significantly altered in both cohorts.

Sex-stratified analyses uncovered dramatically divergent recovery patterns between ME/CFS and control participants (**Fig. 5**). In females, only 21 proteins were significantly altered in the control group, while an extensive 3425 proteins (47% of the dataset) changed significantly in the ME/CFS group. In males, the pattern reversed, with 2544 proteins (35%)

changed significantly in the control cohort, whereas only three proteins changed in the ME/CFS group.

These opposing patterns reveal striking sex-specific differences in the biological response to exertion and recovery. While ME/CFS females showed broad proteomic disruption following Day 1 exercise, ME/CFS males exhibited a blunted or absent response, in sharp contrast to their respective control groups. Although the limited size of the male cohort warrants caution in interpreting our findings, the results suggest distinct underlying pathophysiological processes and highlight the importance of analyzing the two sexes separately in ME/CFS research.

Time-Series C Exposes Cohort-Specific Proteomic Responses to Our Exercise Protocol

To identify coordinated proteomic changes in response to exertion, we applied time-series clustering to significantly altered proteins in any of the six pairwise timepoint comparisons from the linear mixed-effects model (**Fig. 5** and **Supplemental Fig. S3**, $q < 0.05$). For each significant protein, we calculated the median abundance per time point, scaled the values, and performed fuzzy c-means clustering to soft cluster proteins with similar temporal patterns. The optimal number of clusters was determined empirically using the elbow method (41). This analysis was performed separately for the ME/CFS and control groups.

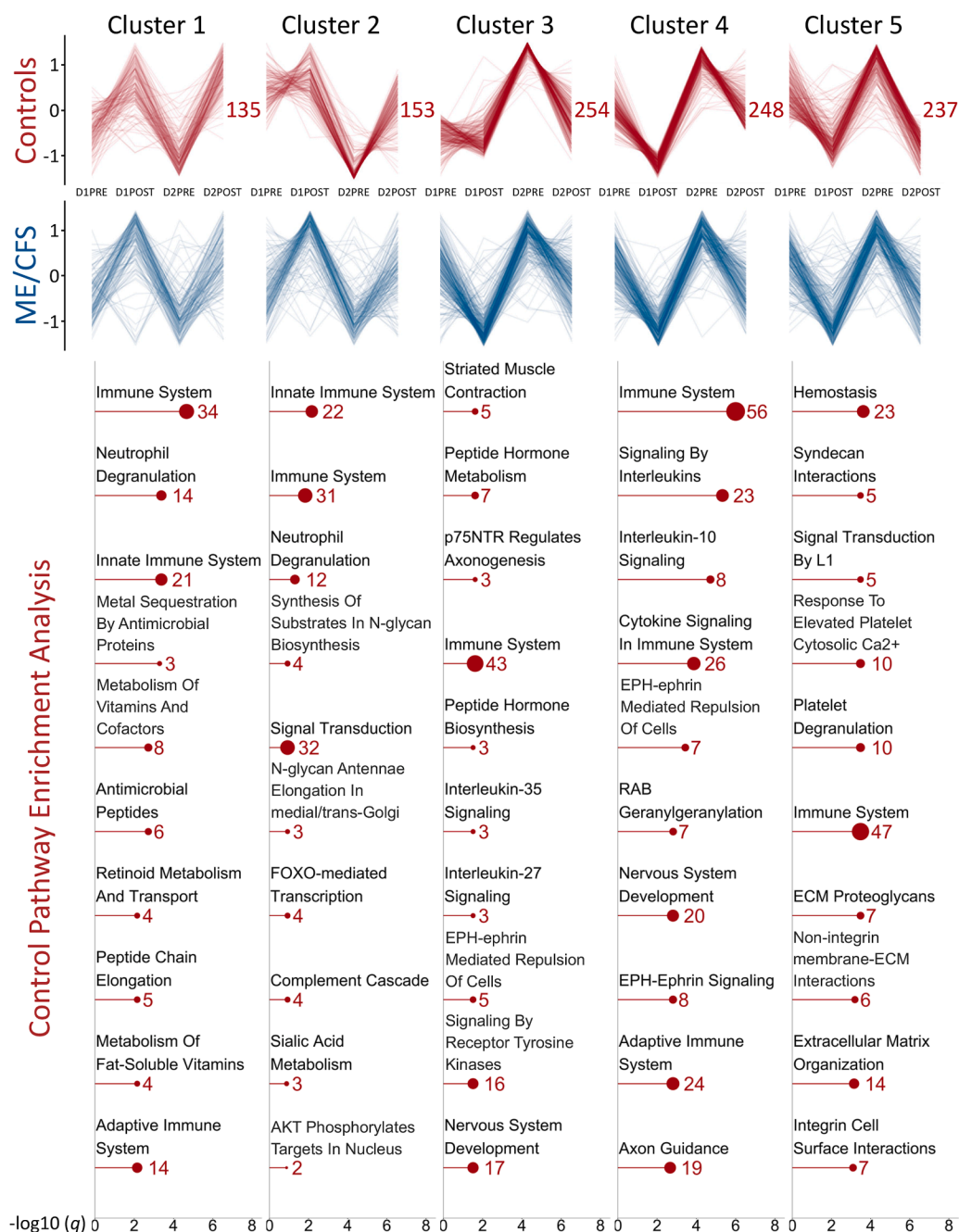


FIG. 6. Fuzzy c-means clustering for controls from the 1027 proteins significantly affected by exercise ($q < 0.05$) and clustered by temporal profiles (red clusters). Each line represents one protein. The numbers next to the red clusters indicate the number of proteins belonging to that cluster. The second row of fuzzy c-means plots (blue) were generated using the same proteins from the ME/CFS cohort. The lollipop plots represent the top 10 Reactome pathways from an enrichment analysis using the proteins from the control clustering graphs (red). The numbers at the end of each lollipop represent the number of proteins within the cluster associated with each pathway from the Reactome database.

The 1027 proteins with significant temporal changes in the total control cohort were grouped into five clusters (red lines, Fig. 6), each representing a distinct temporal response pattern. Cluster 1 features proteins with rising levels during CPET1, decreasing during recovery, and rising again during CPET2. Cluster 5 shows the opposite pattern, while clusters 2

to 4 exhibit intermediate or mixed patterns. Next, we visualized how these same protein clusters behaved in the ME/CFS cohort (blue lines, Fig. 6), regardless of whether they were significantly altered in ME/CFS. While Clusters 1, 4 and 5 showed similar trends between groups (albeit with some proteins with more dispersed trajectories in ME/CFS),

Clusters 2 and 3 displayed marked divergence, with ME/CFS patients showing a larger magnitude of changes during CPET1 compared to controls.

Pathway enrichment analysis of control protein clusters (red lollipop plots, Fig. 6) using Enrichr and the Reactome database (www.reactome.org) revealed that the “immune system” was enriched in all five clusters. Clusters 1 and 2 were enriched for “innate immune system” and “neutrophil degranulation”. Cluster 2 additionally included “complement cascade”, N-glycans necessary for glycosylation and protein folding quality control as well as “signal transduction” (20% of cluster 2 proteins). This pathway, according to Reactome, encompasses processes that alter cell states and activities based on extracellular signals, potentially modulating cellular proliferation, differentiation, and survival throughout the body. Other impacted pathways in cluster 2 include the “FOXO-mediated transcription” pathway that regulates cell survival, growth, differentiation, and metabolism in response to environmental changes. “Sialic acid metabolism” is also implicated in cell signaling. Cluster 3 included numerous signaling pathways, such as interleukin, tyrosine kinases and EPH-ephrin signaling, and was notably enriched for “striated muscle contraction”, implicating both skeletal muscles involved in respiration, locomotion, and posture, and cardiac muscles responsible for blood circulation. Full results are provided in Supplemental Table S4.

We next applied the same pipeline to the 378 proteins significantly changing in the ME/CFS cohort, identifying four optimal clusters (Fig. 7). While ME/CFS protein patterns (blue lines) appeared more cohesive, the control temporal patterns for the same proteins (red) appeared more dispersed, highlighting a distinct response between the two cohorts for many of the proteins.

Clusters 2 and 3 showed strong dysregulation in ME/CFS, with cluster 2 protein levels rising during CPET1 and a more blunted CPET2 response in ME/CFS, contrasting with a more modest response during CPET1 in controls and a stronger increase during CPET2. On the contrary, cluster 3 showed inverted trends in ME/CFS compared to controls.

Pathway analysis for ME/CFS (blue lollipop plots, Fig. 7), showed an enrichment in “immune system”, “integrin cell surface interactions” related to the extracellular matrix and “iron uptake and transport” for cluster 2. Cluster 3 included “EPH-Ephrin signaling” and “Netrin Mediated Repulsion signals”, as well as pathways related B4GALT1 linked to glycosylation (B4GALT1), axonogenesis (p75NTR), and sialic acid metabolism (ST3GAL3). Full results are provided in Supplemental Table S4.

Given the distinct sex-specific patterns seen in earlier analyses, we performed time-series clustering separately for the female ME/CFS cohort (3440 proteins; 47%) and male cohort (2547 proteins; 35%) of our dataset, identifying five clusters in each (Fig. 8). Sample sizes in the female control and male ME/CFS groups were insufficient for reliable clustering.

In females, there were many proteins with divergent patterns in female patients compared to controls (Fig. 8A). Nevertheless, overall patterns showed partial alignment in controls compared to ME/CFS, especially when considering the recovery trajectories (D1POST to D2PRE), suggesting some conserved recovery processes. In males, patients with ME/CFS exhibited proteomic trajectories that were mostly reversed relative to healthy controls across multiple clusters (Fig. 8B), suggesting a sex-specific disruption or inversion in recovery dynamics. While pathway enrichment analysis on the female subgroup primarily contained familiar pathways related to the immune system, cell-cell communication and the extracellular matrix (Supplemental Fig. S4), new pathway enrichments emerged for the male subgroup, with Clusters 1 and 3 highlighting nucleotide metabolism, and cluster 4 enriched in digestive-related processes (Supplemental Fig. S5). While the male results are striking, we remain cautious due to the small size of that cohort. Full results are provided in Supplemental Table S4.

Because pathway enrichment analyses are constrained by the proteins represented in the SomaLogic assay, pathways with larger representation are more likely to appear significant in overrepresentation analyses. To address this limitation, we compared the number of proteins in a given pathway in the full dataset with the number of proteins in enriched pathways in the temporal analysis (Figs. 6–8), with results provided in Supplemental Table S5. Reactome pathway coverage analysis showed that 49% of the 2050 pathways had at least 50% coverage (Supplemental Fig. S6). Large pathways related to signal transduction, the immune system, metabolism, and disease ranged between 30 to 75% coverage, whereas pathways related to olfactory and sensory functions showed little to no coverage (Supplemental Fig. S7). Smaller pathways range from minimal to complete coverage. These results indicate that the SomaLogic assay provides a broad, though uneven, representation of functional pathways in the circulating proteome.

High-Dimensional Reduction in the Proteomics and Metabolomics Datasets From the Same Patients Shows Significant Divergence

We used t-distributed stochastic neighbor embedding (t-SNE) to reduce the dimensionality of the proteomics dataset, enabling visualization of overall sample similarity for each plasma sample specimen (t-SNE plot in Fig. 9). In this two-dimensional space, samples that are more similar in proteomic profile appear closer together. As seen previously with t-SNE applied to metabolomics data from many of the same patients (Fig. 2 in (9)), plasma samples did not group by phenotype or sex in the proteomics dataset (data not shown).

However, when examining sample grouping across time-points for individual participants, we observed a stark contrast between metabolomics and proteomics. In the metabolomics dataset, samples from the same participant

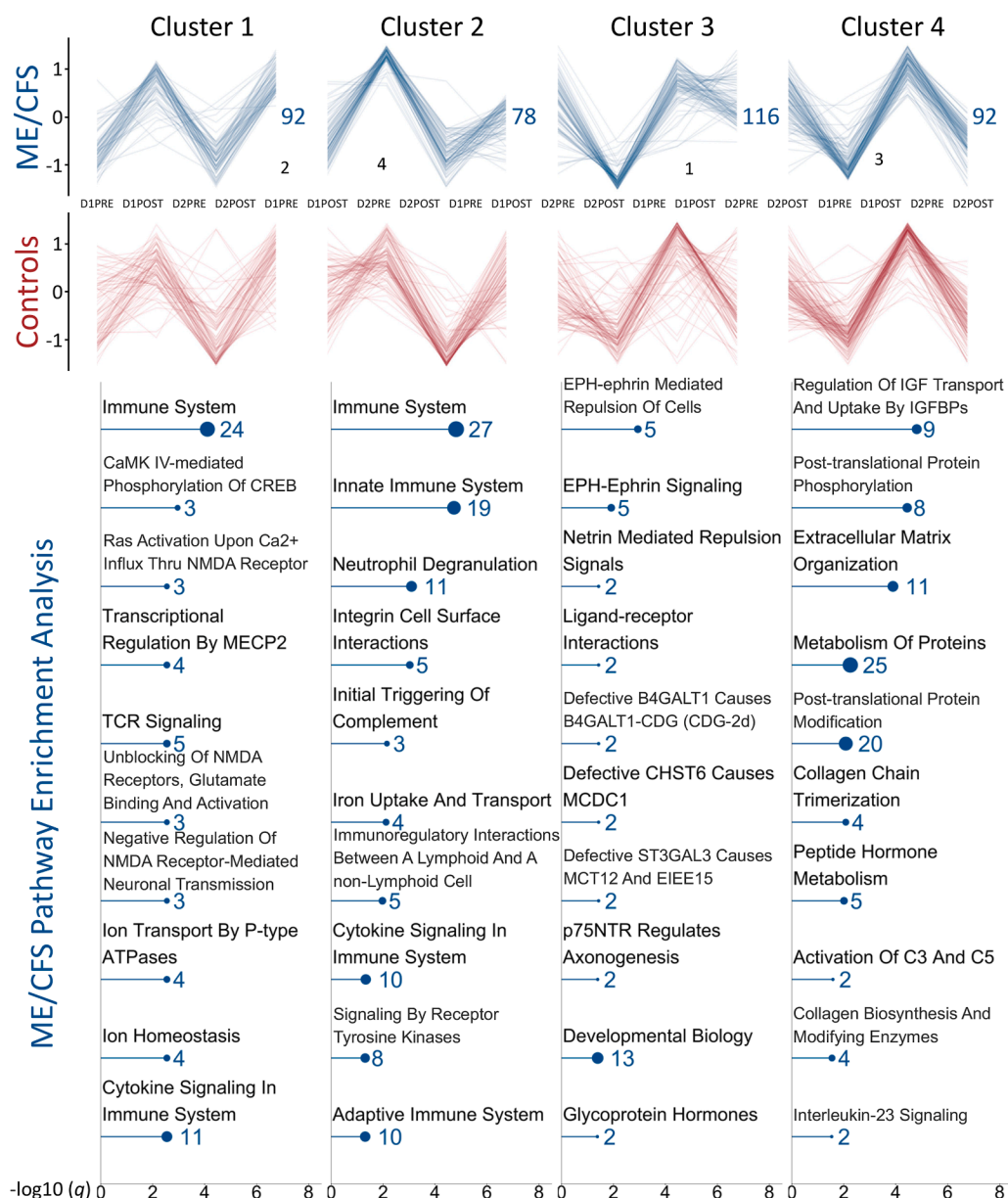


FIG. 7. Fuzzy c-mean clustering for patients with ME/CFS from the 378 proteins significantly affected by exercise ($q < 0.05$) and clustered by temporal profiles (blue clusters). Each line represents one protein. The numbers next to the blue clusters indicate the number of proteins belonging to that cluster. The second row of fuzzy c-means plots (red) were generated using the same proteins from the control cohort. The lollipop plots represent the top 10 Reactome pathways from an enrichment analysis using the proteins from the control clustering graphs (blue). The numbers at the end of each lollipop represent the number of proteins within the cluster associated with each pathway from the Reactome database.

tended to group together across all timepoints. In contrast, only 22 participants showed close grouping of all their timepoints, 62 had one or more timepoints that were dispersed, and 28 had all three timepoints separated (the remaining 20 patients had one or more missing timepoints and could not be conclusively categorized).

To quantify these patterns, we calculated the pairwise distances between the three timepoints for each participant in

t-SNE space. Using a two-component Gaussian mixture model, we identified a cutoff distance of 1.7455, which separated participants into two groups: those whose timepoints remained close in t-SNE space (interpreted as stable or high intra-subject similarity) and those whose timepoints were more dispersed (interpreted as low intra-subject similarity). We refer to these as the high-similarity and low-similarity groups, respectively. This classification was applied

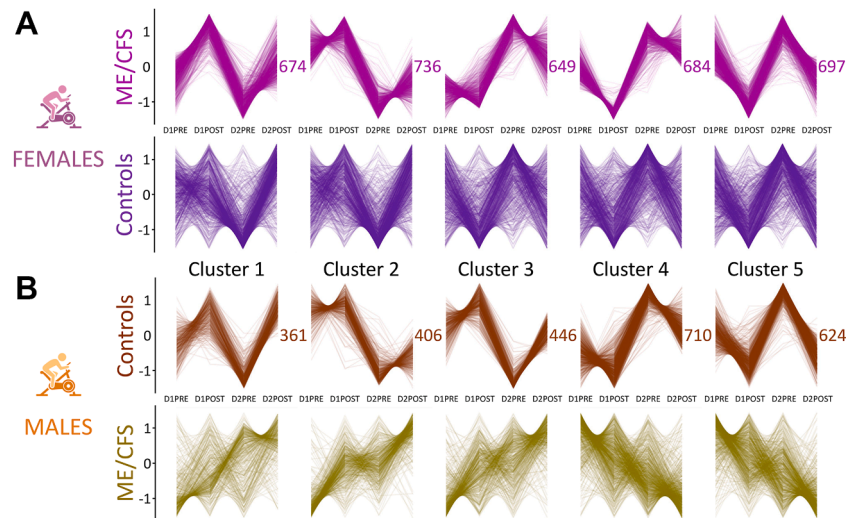


FIG. 8. **Sex-stratified fuzzy c-mean clustering.** A, for female patients with ME/CFS and their 3440 proteins significantly affected by exercise ($q < 0.05$), clustered by temporal profiles (light purple clusters). The second row of clusters (dark purple) were generated using the same proteins from each cluster but for the female control subgroup. B, for male controls and their 2547 proteins significantly affected by exercise ($q < 0.05$) and clustered by temporal profiles (brown clusters). The second row of clusters (olive) were generated using the same proteins from each cluster but for the male ME/CFS subgroup.

independently to each of the three timepoint transitions: Δ Day1, recovery, and Δ Day2, followed by “_HS” for the high-similarity group and “_LS” for the low-similarity group.

Despite extensive exploration of this grouping strategy and applying the same analytic pipelines presented in Figures 2–9, no clear unifying pattern emerged. Nevertheless, we report these intriguing inter-individual differences and present a logistic regression analysis to further investigate potential explanatory variables.

We modeled the relationship between t-SNE-derived group assignments (Fig. 9) and a range of demographic, clinical, and exercise physiology parameters (Table 1) using logistic regression. Detailed model outputs, including regression

coefficients, p -values, and odds ratios for all predictors across the specified timeframes, are provided in Supplemental Table S6. In this analysis, odds ratios >1 indicate a higher likelihood of assignment to group “_HS”, while odds <1 reflect a higher likelihood of assignment to group “_LS”.

Although phenotype (ME/CFS or control), general health, and sex were not significantly associated with group assignment at any investigated timeframe (Table 1), several other factors emerged as noteworthy. Age was a significant predictor at Δ Day1, with an odds ratio of 1.52 per 10-year increase. Similarly, BMI showed a significant effect at Δ Day2, with an odds ratio of 1.58 per 5 kg m⁻² increase.

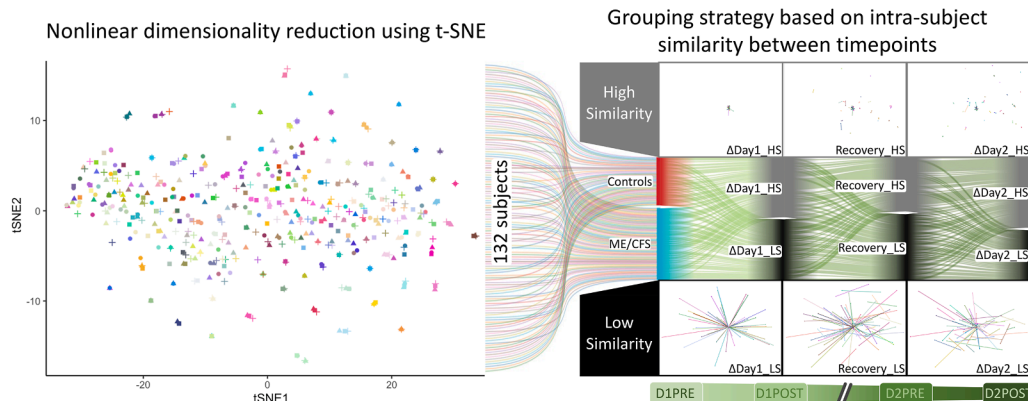


FIG. 9. **Grouping strategy derived from the t-SNE nonlinear dimension reduction.** Each shape in the t-SNE plot represents (D1PRE as triangles, D1POST as circles, D2PRE as squares and D2POST as crosses) and their color is specific to each subject. High intra-subject similarity refers to distance less than 1.7455 while low intra-subject similarity is higher than 1.7455. The grouping was done for changes occurring as a result from the impact of exercise on Day1 (Δ Day1), during recovery (Recovery) and as a result from the impact of exercise on Day2 (Δ Day2).

TABLE 1

Odds ratios for selected predictors including demographic, clinical, and exercise physiology parameters, for Δ Day1, Recovery and Δ Day2 – p-values from the Wald test are shown in bold for $p < 0.05$ and are omitted for $p > 0.15$

Predictors	Odds ratio		
	Δ Day1	Recovery	Δ Day2
Phenotype	0.38	0.42	5.36 ($p = 0.11$)
General Health	0.98	0.98	1.02
Sex	0.75	1.51	0.82
Age (10 years)	1.52 ($p = 0.03$)	1.05	1.07
BMI (5 kg/m ²)	0.86	0.98	1.58 ($p = 0.049$)
VO ₂ max1	1.02	0.97	1.09 ($p = 0.07$)
VO ₂ max2	1.02	0.99	1.13 ($p = 0.009$)
AT1	1.02	0.99	1.12
AT2	0.98	0.98	1.23 ($p = 0.02$)
LA versus Ithaca	3.14 ($p = 0.02$)	3.67 ($p = 0.02$)	1.51
NYC versus Ithaca	1.09	1.15	0.72
NYC versus LA	0.35 ($p = 0.06$)	0.31 ($p = 0.047$)	0.48

LA, Los Angeles; NYC, New York City.

These findings suggest that older individuals and those with higher BMI are more likely to experience greater divergence across timepoints in proteomic space, as captured by t-SNE. Additionally, cardiorespiratory fitness measures, specifically VO₂max2 and AT2 were significantly associated with group assignment at Δ Day2, with odds ratios of 1.13 and 1.23 mL kg⁻¹.min⁻¹, respectively. This indicates that greater cardiorespiratory fitness is linked to a higher likelihood of low intra-subject similarity in proteomic profiles after the second post-exercise timepoint. Test site also emerged as a significant factor. Participants tested in Los Angeles, compared to Ithaca, were 3.14 times more likely to be assigned to group “_LS” at Δ Day1, and 3.67 times more likely during recovery. Additionally, the New York City versus Los Angeles comparison reached significance during the recovery phase. These site-specific differences reinforce the importance of adjusting for test location in all our regression models, a step we have consistently implemented throughout our analyses.

The significant effect of age, BMI and site on intra-individual longitudinal exercise profiles highlights the importance of adjusting analyses for these confounding factors and comparing ME/CFS patients to sedentary controls, which we have done in this study, rather than to healthy controls with much higher cardiovascular fitness. While associations with age, BMI, VO₂max2 and AT2 are statistically significant, the odds ratios are moderate. This suggests that other unmeasured factors, potentially biological or behavioral, may influence the observed variability in proteomic response to exercise. Nonetheless, these results showcase the complexity of the proteomic response to exercise and factors that influence this response, even among controls.

Molecular Predictors of Cardiovascular Fitness Using VO₂max and Anaerobic Threshold

To identify proteomic signatures linked to cardiovascular fitness, we performed linear regression analyses modeling

VO₂max and AT as functions of plasma protein levels at each of the four timepoints, adjusting for confounders. This method allowed us to quantify relationships between molecular markers and exercise capacity in both ME/CFS patients and sedentary controls. Although both groups completed the same CPET protocol, the molecular predictors of cardiovascular fitness differed markedly in consistency and pattern (Fig. 10 and Supplemental Table S7).

In the ME/CFS cohort, levels of several proteins emerged as consistent predictors of VO₂max. Most notably, HNF4A was negatively associated with both VO₂max1 and VO₂max2 at all timepoints (Fig. 10, A and B). This transcription factor, essential for hepatic metabolism and circadian regulation (42, 43), may reflect impaired metabolic coordination in ME/CFS. Four additional proteins, GHR, IGFBP2, INHBC, and leptin (LEP), were associated with VO₂max2 at all timepoints, suggesting they are independent of acute exertion and may reflect underlying differences in the general cardiovascular fitness of patients with ME/CFS.

Beyond these consistent predictors, numerous proteins were associated with VO₂max at specific timepoints, revealing dynamic molecular responses. At D1PRE, 18 proteins were significantly associated with VO₂max1, with eight unique to this timepoint (Fig. 10A). Positive associations with ANTXR2, DMBT1, and EPHA3 suggested roles for cell signaling and immune regulation in baseline exercise capacity. Conversely, negative associations with ITGB2, KMT2C, OLFML3, and VTN implicated chromatin remodeling, immune response, and extracellular matrix interactions as potential contributors to impaired performance. Of these, the association with ANTXR2 was not specific to ME/CFS, as it was also significant in controls.

Following the first CPET, at D1POST, a broader molecular response was evident. A total of 33 proteins were associated with VO₂max1, 23 of which were exclusive to this timepoint (Fig. 10A). These included proteins involved in cellular

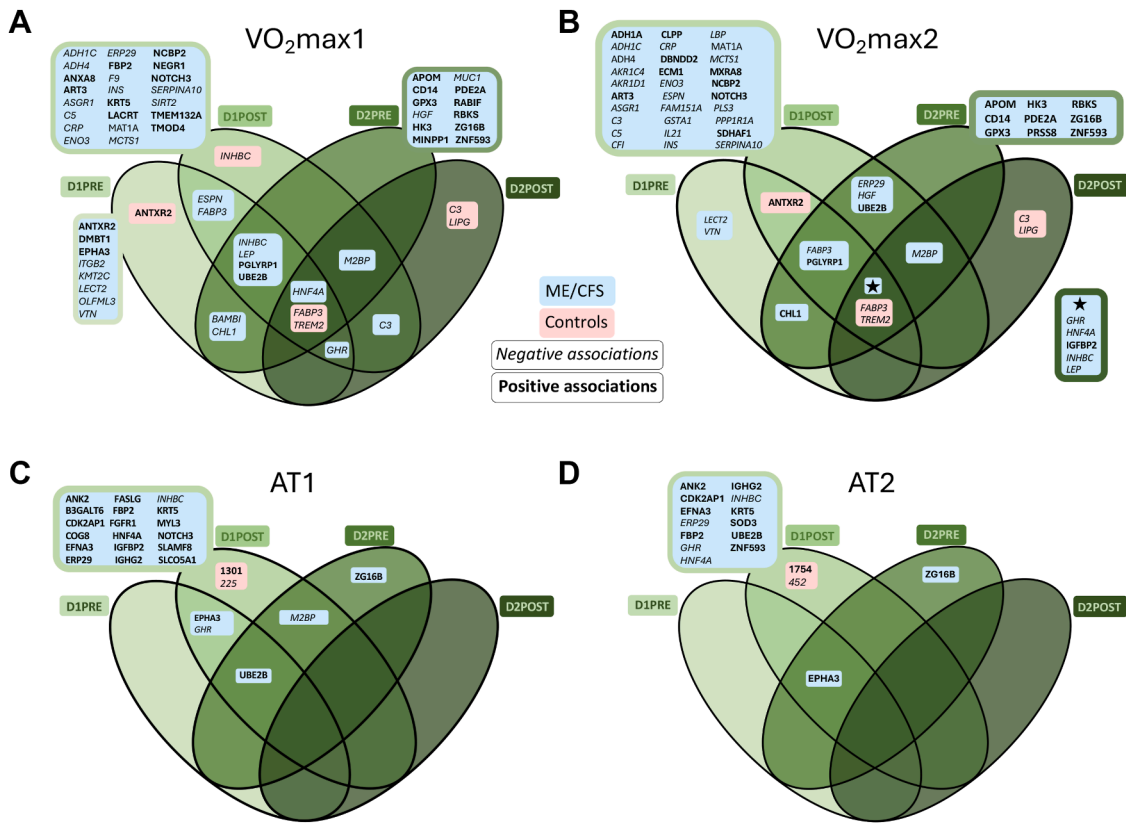


FIG. 10. **Protein associations with CPET parameters at each timepoint in patients with ME/CFS and healthy controls.** Venn diagrams showing statistically significant proteins associated with (A) VO₂max 1; (B) VO₂max 2; (C) anaerobic Threshold 1 (AT1); and (D) anaerobic threshold 2 (AT2), at $q < 0.1$. Green-shaded sections indicate the timepoints D1PRE, D1POST, D2PRE, and D2POST. Blue boxes contain proteins statistically significant in ME/CFS patients, and pink boxes indicate proteins statistically significant in healthy controls. Proteins positively associated are displayed in bold, while those negatively associated are in *italics*.

signaling and neuroregulatory processes (ANXA8, ART3, NCBP2, NEGR1, TMEM132A), glucose metabolism (FBP2), and muscle physiology (TMOD4). Negative associations were observed with a cluster of proteins related to muscle energy metabolism (ENO3), metabolic regulation (INS and MAT1A), coagulation processes (SERPINA10 and F9), and stress response (SIRT2). These patterns likely reflect acute inflammatory and metabolic strain post-exertion. For VO₂max2 (Fig. 10B), 41 proteins were associated at D1POST, 30 of which were unique. Among them, ECM1 and MXRA8, involved in extracellular matrix remodeling, may contribute to vascular and tissue recovery, while SDHAF1 supports mitochondrial function. Elevated levels of proteins related to inflammation and immune response (CRP, C3, C5, and IL-21) were negatively associated, suggesting that a pronounced inflammatory response may limit post-exercise performance in ME/CFS.

At D2PRE, a timepoint corresponding to the onset of PEM in many ME/CFS participants, 20 proteins were associated with VO₂max1, 12 of which were unique (Fig. 10A). Positive associations with GPX3 and HK3 pointed to improved oxidative stress handling and glucose metabolism in patients

with better cardiovascular fitness, while negative associations with HGF (tissue repair) and MUC1 (cellular protection) suggested continued physiological stress or incomplete recovery. For VO₂max2 (Fig. 10B), 21 proteins were significantly associated, including HK3, which again emerged as a favorable predictor of fitness.

By D2POST, few associations remained, with only four proteins linked to VO₂max1 (M2BP, HNF4A, GHR, and C3), and six with VO₂max2, none of which were unique to this timepoint.

In contrast to the fluctuating patterns in ME/CFS, protein associations with VO₂max in controls were more stable across timepoints and both CPETs (Fig. 10, A and B). This may be due in part to more stable performance on the CPET in controls, whereas patients with ME/CFS demonstrated reduced VO₂max following the second CPET (Fig. 1). FABP3 and TREM2, involved in lipid metabolism and immune response, respectively, were negatively associated with VO₂max across all time points and both CPETs. TREM2 was not a significant predictor in any of the ME/CFS results, indicating dysregulation between levels of this protein and cardiovascular fitness in ME/CFS patients. C3 and LIPG were

negatively associated with $\text{VO}_{2\text{max}}$ at D2POST, potentially reflecting immune and endothelial recovery limitations for patients with lower cardiovascular fitness.

We also examined predictors of AT in patients with ME/CFS (Fig. 10, C and D), with a modest number of proteins associated with AT1 and AT2 at D1PRE, including positive associations with EPHA3 and UBE2B, involved in neuroimmune signaling and cellular stress responses. The post-exercise timepoint D1POST revealed a larger molecular signature with 22 proteins linked to AT1, and 14 to AT2. 18 of the 22 proteins associated with AT1 were positively associated, including FASLG and SLAMF8 (immune activation), as well as structural markers such as ANK2, MYL3, and KRT5. At D2PRE, the positive association of ZG16B, may indicate that higher levels are associated with improved gut mucosal or immune recovery in more fit patients. For AT2, 14 proteins were significantly associated at D1POST, with positive associations for 9 proteins linked to structural integrity (ANK2, KRT5), metabolic regulation (FBP2, CDK2AP1), oxidative stress response (SOD3), transcriptional regulation (ZNF593, UBE2B), cellular signaling (EFNA3), and immune functions (IGHG2). Negative associations were found for four involved in inflammatory responses (INHBC, ERP29), metabolic regulation (GHR), and transcriptional control (HNF4A).

In the control group, AT predictors were abundant, with 1526 proteins associated with AT1 at D1POST (1301 positively), and 2206 with AT2 (1754 positively). This large set of post-exercise associations in controls points to a robust and coordinated physiological response to acute exercise that varies by fitness level. Enrichment analysis using Enrichr revealed that approximately 20% of these proteins were involved in key Reactome pathways, including immune system signaling, cytokine regulation, and interleukin pathways (Supplemental Table S8), highlighting systemic immune involvement in regulating exercise capacity.

Correlation of Protein Levels With Clinical Parameters in Patients With ME/CFS

We evaluated Spearman correlations between plasma protein levels and clinical parameters, including age, BMI, SF-36 general health score (GH), SF-36 physical component summary (PCS), specific symptom severity (SSS) scores, and the multidimensional fatigue inventory (MFI)-20. Symptoms were assessed at baseline and post-exercise timepoints, allowing both static and dynamic (delta) comparisons. Delta scores captured changes in symptom severity from D1PRE to D2PRE (Delta1), from D1PRE to 2 days post CPET 1 (Delta2), and from D1PRE to 4 days post CPET 1 (Delta3), with positive values indicating symptom worsening and negative values indicating symptoms improved. Proteins showing moderate correlations ($|R| > 0.5$ and $q < 0.1$) are highlighted below with full results in Supplemental Table S9.

In the total ME/CFS cohort, several proteins correlated with clinical status. For instance, higher D1PRE levels of SHISA2

(Shisa family member 2), a myoblast fusion regulator (44), correlated with better GH, unlike for controls (Fig. 11A). D2PRE levels of BAGE3 (B melanoma antigen family 3, a protein of unknown function) were inversely correlated with MFI-20 total score at D2PRE (Fig. 11B). At D2POST (Fig. 11C), decreased levels of CRPPA (CDP-L-ribitol pyrophosphorylase A, involved in nucleotide synthesis), NCAM1 (Neural cell adhesion molecule 1), and PRPS2 (Phosphoribosyl pyrophosphate synthetase) were linked to worsening of muscle pain and lymph node tenderness post-exercise and more severe recent PEM, respectively. CRPPA (also known as ISPD) plays a role in the synthesis of glycosylation precursors and is essential for muscle fiber stability in poultry (45). NCAM1 is a cell adhesion molecule involved in synaptic plasticity and immune interactions. Its reduction in patients with more severe lymph node tenderness following the 2-day CPET may indicate disrupted neuro-immune interactions which may be contributing to or reflective of the overall state of immune dysregulation in ME/CFS patients during the recovery from exercise (46). PRPS2 is critical for nucleotide biosynthesis and lower levels at D2POST could imply impaired energy metabolism, analogous to its well-established role in cellular stress responses (47). Thus, patients with more severe recent PEM prior to the exercise challenge may have been pre-disposed to impaired energy metabolism post-exercise.

We also assessed correlations between within-subject changes in protein levels and symptom dynamics. A greater exercise-induced drop in SERPINB1 (Serpin family B member 1) levels on Day1 correlated with worse GH scores (Fig. 11D). SERPINB1 plays an essential role in the regulation of the innate immune response, inflammation, and cellular homeostasis. Changes in TNFSF4 (TNF superfamily member 4 protein) levels during recovery were linked to reduced SF-36 PCS scores (Fig. 11E). TNFSF4 is a cytokine recognized for its role in co-stimulating T-cell proliferation and cytokine production. DBI (Diazepam binding inhibitor) levels due to Day2 exercise were associated with increased sore throat severity (Fig. 11F). DBI is an acyl-CoA binding protein that is regulated by hormones and plays a role in lipid metabolism. Positive shifts in MICU3 (Mitochondrial calcium uptake family member 3) and NDRG3 (NDRG family member 3) levels over the full protocol correlated with better GH (Fig. 11G). MICU3 is a tissue specific enhancer of mitochondrial calcium uptake, while NDRG3 (NDRG family member 3), is predicted to be involved in signal transduction. Decreasing levels of OXLD1 (Oxidoreductase like domain containing 1, a protein with unknown function) tracked with reduced sore throat symptoms (Fig. 11H). These findings highlight associations between immune signaling, mitochondrial regulation, symptom and overall disease severity in ME/CFS patients. All of these correlations were very weak or absent in controls.

In female ME/CFS participants, 154 protein-symptom correlations passed the significance threshold ($|R| > 0.5$ and

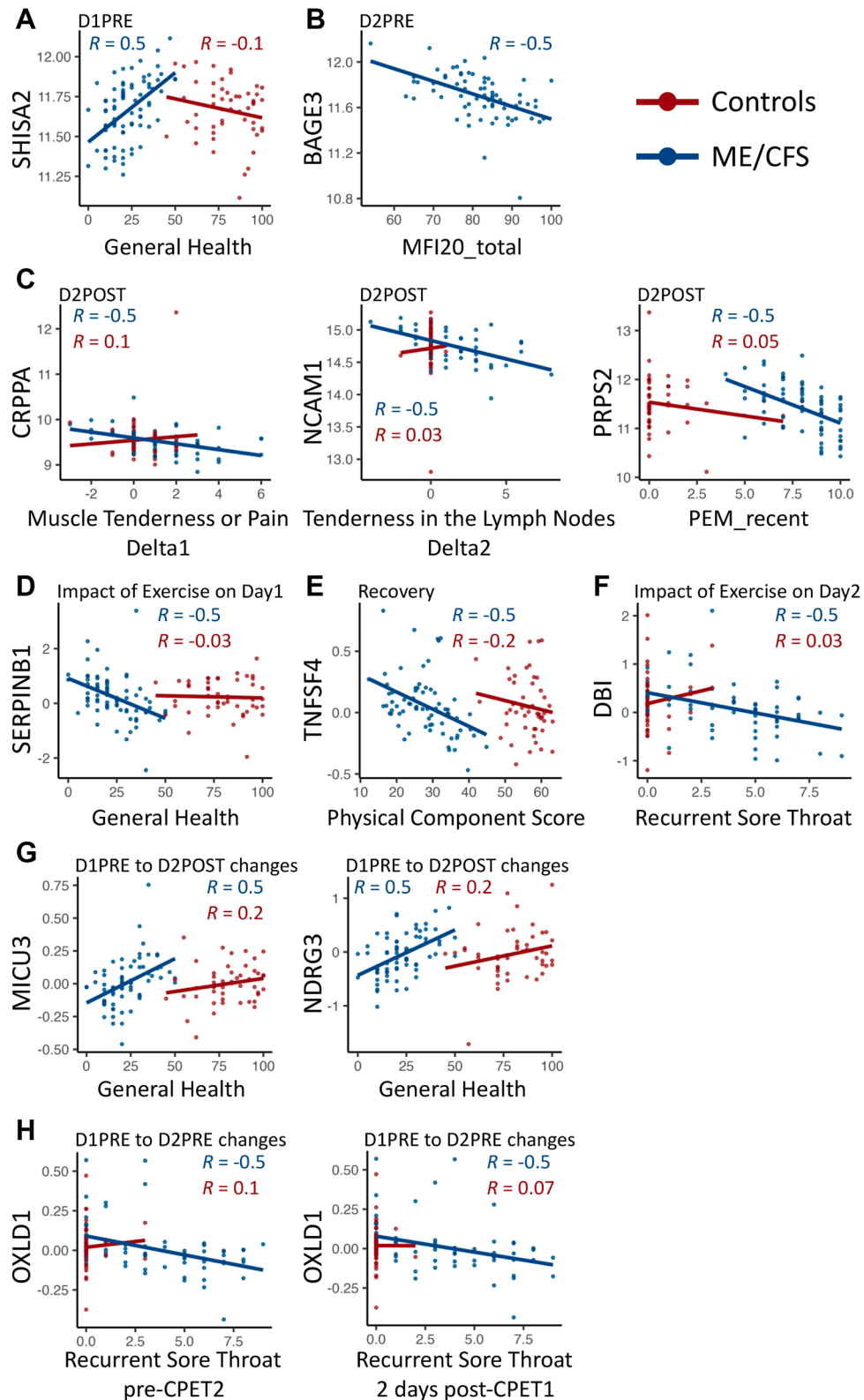


FIG. 11. **Significant Spearman correlations with demographic, clinical, and exercise physiology parameters, for the total cohort.** Linear regression lines and Spearman's R are shown on each plot for controls (red) and ME/CFS patients (blue). The SSS scores range from 0 when the symptom is absent to 10 for very severe. For SF-36 general health and physical component score, a higher score indicates better health. For the MFI-20 total score, a higher number indicates worse overall fatigue. Controls did not complete the MFI-20 questionnaire. All correlations presented are significant in ME/CFS patients at $q < 0.1$ and $|R| > 0.5$.

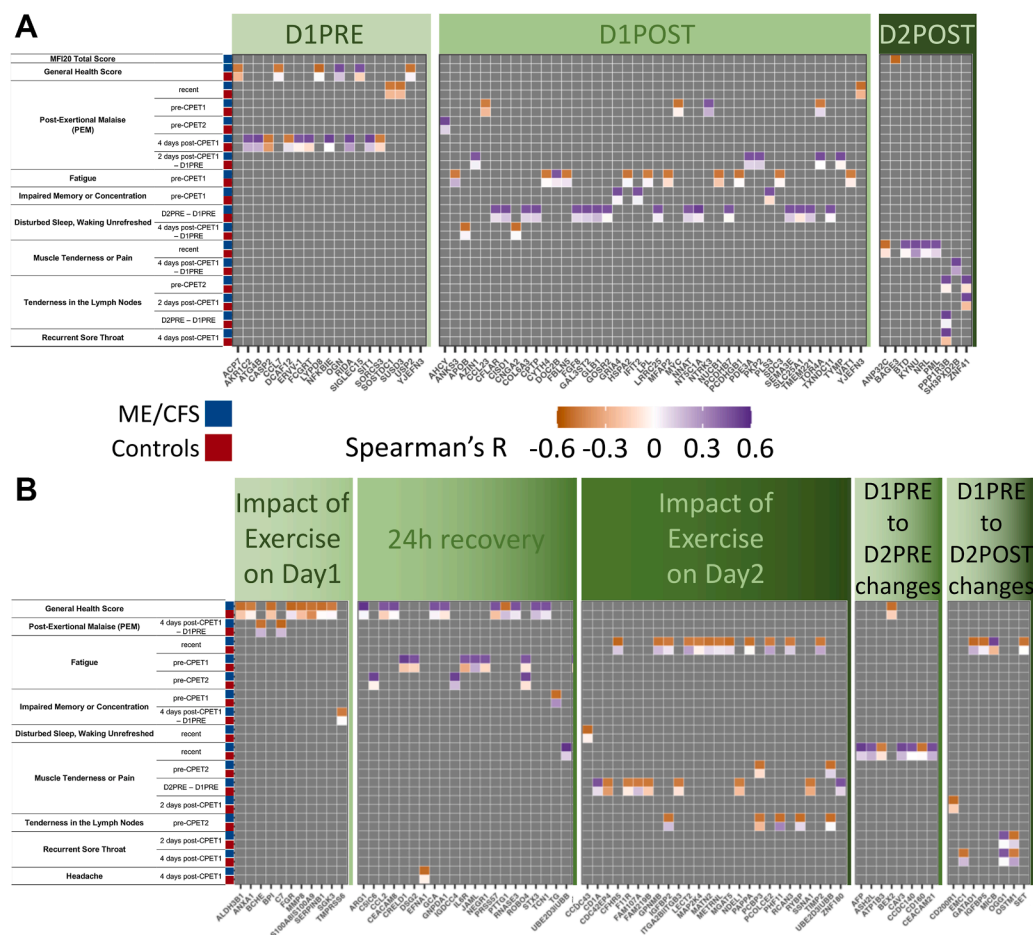


FIG. 12. Heatmap of significant correlations with clinical data for the ME/CFS female cohort. Each heatmap panel shows the Spearman correlation coefficients between proteins and clinical parameters at (A) individual timepoints or (B) for changes over time (B) for $|R| > 0.5$ and $q < 0.1$. Within each clinical parameter, the R values for the comparisons that did not meet the criteria are not shown (gray boxes). The table describes the clinical parameters.

$q < 0.1$). Figure 12A shows the 74 significant correlations across the single timepoints, while Figure 12B shows the 80 significant correlations across timepoint ratios. Recent PEM severity was negatively associated with SOSTDC1 (Sclerostin domain containing 1) and SUSD3 (Sushi domain containing 3) at D1PRE, and YJEFN3 (YjeF N-terminal domain containing 3) at D1POST (Fig. 12A). SOSTDC1 is a bone morphogenetic protein (BMP) antagonist, which is crucial for cell growth, apoptosis, and differentiation. YJEFN3 regulates angiogenesis by accelerating cholesterol efflux from endothelial cells to high-density lipoprotein. Negatively and positively associated changes in 19 proteins across D1PRE to D1POST, the recovery period, and D1PRE to D2PRE, tracked with GH scores (Fig. 12B), reflecting both beneficial and detrimental molecular responses to exertion. Changes in SERPINB1 MMP8 (Matrix metalloproteinase 8) levels following CPET1 were negatively associated with GH, implicating immune modulation and extracellular matrix remodeling (Fig. 12B).

Recent average fatigue in female ME/CFS patients was linked to changes in 16 proteins across late exercise intervals, with 15 of these showing negative correlations. MICB (MHC class I polypeptide-related sequence B), a stress-induced immune ligand, was the only protein positively associated with recent fatigue from D1PRE to D2POST.

To explore the biological relevance of these associations at D1PRE, we conducted Reactome pathway enrichment using Enrichr (21), for proteins significantly correlated with GH, sore throat, and unrefreshing sleep in the female ME/CFS cohort ($q < 0.05$ significance threshold). GH-associated proteins ($n = 230$) were enriched for interleukin and cytokine signaling, immune system, angiogenesis and extracellular matrix organization pathways. For recurrent sore throat ($n = 247$), enrichments included various hits for the immune system, cytokine and interleukin signaling, and angiogenesis pathways related to vascular endothelial growth factor. The only significant pathway for unrefreshing

sleep ($n = 61$) was “Plasma Lipoprotein Assembly, Remodeling, and Clearance”, suggesting a metabolic link to sleep disturbance.

In summary, multiple immune, metabolic, and structural proteins correlated with symptom burden and recovery dynamics in ME/CFS, particularly among female participants. These findings support the hypothesis that post-exertional symptom fluctuations as well as overall disease severity are mirrored by specific, biologically relevant shifts in circulating protein levels.

DISCUSSION

Our large-scale study provides a comprehensive profile of the plasma proteome in individuals with ME/CFS and in matched sedentary controls. To distinguish disease-specific molecular signatures from those related to the physical deconditioning often observed in patients with ME/CFS, we carefully selected sedentary controls matched for age and BMI. Importantly, our study also offers novel insights into exercise-induced molecular responses in a predominantly female sedentary control population, a group historically underrepresented in exercise physiology research. Although ME/CFS was our primary focus, the pathway analysis in the control group (Fig. 6) reveals dynamic plasma proteomic changes following two acute exercise bouts, particularly in various immune responses and cell-cell communication pathways.

With a cohort of 132 patients (Fig. 1B) and the use of the enhanced SomaScan 7K Assay, which quantifies proteins across a dynamic range of 10 orders of magnitude, we achieved unprecedented depth of baseline plasma proteomic profiling in ME/CFS and matched controls. SomaScan technology has previously been applied in 2 ME/CFS studies, including our own earlier work utilizing the 5K Assay to analyze 5284 SOMAmers representing 4790 unique proteins (15). That study identified significant disruptions in Ephrin-Eph and immune system signaling between patients and controls, consistent with findings in the current study. In contrast, another investigation using the SomaScan 1.3 K Assay (1305 analytes) on a smaller cohort failed to detect significant proteomic differences (16). In our current analysis, no proteins were significantly different at baseline after stringent statistical FDR correction (Figs. 2 and 3). This contrasts with our prior pilot study, which applied a less stringent statistical approach (Wilcoxon rank-sum test with no confounder adjustments) and identified nine proteins elevated in ME/CFS at baseline, with $q < 0.05$ (15). The discrepancy underscores the importance of statistical power and appropriate correction methods when interpreting high-dimensional proteomic data particularly in relation to sample size (40 vs. 132 patients across studies) and when examining subgroups such as males. Notably, several other proteomics studies have reported baseline differences using alternative detection

platforms, but without applying comparable multiple testing corrections (48–50).

Traditionally, protein identification and quantification have relied on targeted techniques such as enzyme-linked immunosorbent assay (ELISA) of a few proteins, or liquid chromatography coupled with mass spectrometry (LC-MS or LC-MS/MS) for broader, untargeted profiling. Both approaches have been applied in ME/CFS research (51–55), including studies examining the protein cargo of circulating extracellular vesicles (56–60). Across these investigations, immune dysregulation has emerged as a consistent feature in ME/CFS, including elevated inflammatory markers (61, 62) and disrupted T cell metabolism (63–66). Our baseline GSEA reveals compelling differences in intercellular signaling in ME/CFS cases, notably in “T cell receptor signaling” and other mechanisms of cell-cell communication (Fig. 4). These findings align with broader evidence from ME/CFS research, including a recent single-cell transcriptomics study that identified altered immune signaling across multiple cell types including T cells (67). They are also supported by our previous proteomics work (15) and an independent MS/MS study, both of which highlighted immune involvement, particularly B-cell dysfunction (54).

Although exercise is widely promoted for maintaining physical health and managing numerous chronic conditions (4–6, 68–71), it can be detrimental to individuals with ME/CFS by triggering PEM. Acute bouts of exercise induce complex molecular responses that reflect the body’s capacity to adapt to physiological stress (72) and longitudinal sampling in animal and human studies has been crucial in unraveling these adaptive changes (68, 72–74). A strength of our study is its longitudinal sampling protocol involving two acute exercise sessions separated by a 24-h recovery period (Fig. 1A), deliberately designed to elicit PEM in ME/CFS participants (17). This approach enhances our ability to monitor dynamic changes in the plasma proteome, reduces inter-individual variability, and increases statistical power for detecting group differences.

Our most compelling findings emerged from the analysis of temporal shifts in the plasma proteome, (Figs. 5–9, 11 and 12, along with Table 1). Approximately 15% of proteins in the control group showed significant alterations during the 24-h recovery window, compared to only 5% in the ME/CFS group (Fig. 5). Clustering analysis (Figs. 6 and 7) revealed distinct pathway perturbations in response to exercise, including those related to immune system biology, Ephrin-Eph signaling, cell-cell communication pathways, and muscle physiology. These results align with prior ME/CFS research (9, 15, 54, 63, 65, 67, 75) and underscore the widespread and systemic dysregulation that characterize PEM in ME/CFS.

The D2PRE timepoint (24 h of recovery after the first bout of exercise and coinciding with the onset of PEM) exhibited the greatest proteomic dysregulation in ME/CFS compared to

controls (Figs. 2–4). Notably, “glycolysis/gluconeogenesis” was the only pathway upregulated in patients in the GSEA (Fig. 4), a finding consistent with prior studies reporting impaired oxidative phosphorylation and increase reliance on anaerobic metabolism (9, 51, 70–72). This shift may reflect a compensatory response to mitochondrial stress, potentially contributing to early fatigue, lactate accumulation, and PEM. At this same timepoint, linear modeling identified five proteins that differed significantly between ME/CFS and controls, with three involved in muscle physiology (Figs. 2 and 3). Three immune-related pathways, “T cell receptor signaling,” “B cell receptor signaling,” and “IL-17 signaling,” were all significantly downregulated post-exercise in ME/CFS (Fig. 4). These pathways are central to adaptive immunity, and their suppression suggests impaired immune cell activation, communication, and cytokine production. This observation aligns with prior evidence of T cell metabolic dysfunction and impaired NK cell cytotoxicity in ME/CFS (63–67). Notably, the persistence of these immune perturbations into the recovery phase suggests a failure to resolve immune activation, which may underlie the prolonged recovery and immune flares that define PEM. Several neurologically relevant pathways also emerged from our analysis, namely “axon guidance,” “dopaminergic synapse” and “inflammatory mediator regulation of TRP channels” (Fig. 4). These may relate to commonly reported neurological symptoms in ME/CFS such as cognitive impairment, sensory hypersensitivity, and pain. Transient receptor potential (TRP) channels, in particular, are ion channels broadly expressed across tissues and involved in key physiological processes, such as pain sensation, thermo-sensation, neurotransmitter and neuropeptide release, mechanotransduction, vasodilation, and immune function. Dysfunction of TRPM3, a TRP channel subtype, has been previously reported in NK cells from ME/CFS patients, leading to reduced calcium influx (76, 77). A recent review suggests that TRPM3 dysfunction may contribute to immunological disturbance, skeletal muscle impairment, and neurological symptoms in ME/CFS (78). Moreover, an *in vitro* study indicates that naltrexone, which is sometimes prescribed to ME/CFS patients, may help restore TRPM3 function (79).

Sex-specific analyses further emphasized the importance of considering sex as a biological variable in ME/CFS research, particularly when analyzing the temporal dynamics of the proteomic response to acute exercise. However, interpretation of results from the male cohort should be approached with caution due to the limited sample size. The volcano plots in Figure 5, further explored in Figure 8 and Supplemental Figs. S4 and S5, reveal sex-specific alterations in the proteomic response to acute exercise, impacting pathways across a wide range of biological processes. Sex-dependent changes in patients undergoing an exercise challenge to induce PEM have been studied at the

transcriptional level at baseline, peak exercise and 4 hours post-exercise (80, 81). These investigations reported activation of immune-cell signaling and NK cell cytotoxicity in a 13-male ME/CFS cohort, whereas no significant stress-induced transcriptional changes were detected in a 20-female ME/CFS cohort. Although these findings appear to contrast with our proteomic data (Fig. 5), it is important to note that transcriptomic and proteomic responses may occur on different timescales and reflect distinct regulatory layers of the biological response. Given that ME/CFS predominantly affects females, most studies, including ours, enroll more female than male participants (9, 16, 82). Similarly, in our metabolomics analysis of the same plasma samples, we observed sex-related differences. However, due to the small number of male patients, we were conservative in the interpretation and reporting of those findings (9). We also completed sex-stratified analyses of proteomic cargo in extracellular vesicles in female and male cohorts, and while we found some overlap, the significant alterations between ME/CFS patients and controls were predominantly very different in males and females (57, 60).

Our exploration of protein associations with exercise physiology parameters, specifically VO_2max and anaerobic threshold, along with various clinical measures before and during exercise that showed symptom progression, identified numerous potential molecular contributors to ME/CFS symptomatology, including PEM (Figs. 10–12). In a prior study using the SomaScan 5K Assay, linear regression of baseline VO_2max and protein levels in a cohort of 650 sedentary adults undergoing an endurance-exercise intervention identified 147 significant associations (83). Remarkably, three of the six proteins associated with VO_2max in our much smaller control cohort ($n = 53$) overlapped with that study: ANTXR2 showed a positive association, while FABP3 and INHBC were negatively associated in both datasets. In contrast, the ME/CFS cohort in our study showed a larger number of protein- VO_2max associations (Fig. 10, A and B) with reduced consistency compared to the control cohort between $\text{VO}_2\text{max}1$ and $\text{VO}_2\text{max}2$, suggesting a maladaptive response to acute exercise and reflecting the inability of most ME/CFS patients to repeat their CPET performance 2 days in a row. Additionally, proteins MYBPC1 and MYL3, which were positively associated with baseline VO_2max in the larger prior study (Fig. 3 of (83)), did not replicate this relationship in our dataset. However, both proteins showed significantly reduced levels in patients with ME/CFS compared to sedentary controls at D2PRE (recovery) (Fig. 3).

Correlations between protein levels and clinical data of the ME/CFS cohort encompassed hallmark symptoms of the disease, including PEM, muscle tenderness, fatigue and immune-related symptoms such as lymph nodes tenderness and sore throat (Figs. 11 and 12). Many of the symptom-associated proteins were immune-related proteins and

correlated with immune-mediated symptoms, further strengthening the evidence that ME/CFS patients exhibit impaired regulation of inflammation following acute exercise stress (61–67).

A central challenge in ME/CFS research remains the marked heterogeneity of the patient population, influenced by factors such as illness duration, genetic background, environmental exposures, and other factors (84). Our statistical framework anchored in longitudinal sampling, helped address inter-individual variability and revealed consistent disease-relevant signals in ME/CFS. Across timepoints and analyses, we observed a convergence of immune dysregulation, impaired cell-cell communications, and abnormalities in energy metabolism and muscle physiology, hallmarks of a maladaptive response to physiological stress. These alterations were particularly pronounced during the recovery period, when PEM has been triggered in patients with ME/CFS. Notably, we revealed sex-specific differences in the proteomic response to exercise, underscoring the need to consider sex as a strong biological variable in ME/CFS research. By combining deep proteomic profiling with dynamic temporal analyses, our study highlights the systemic nature of ME/CFS pathophysiology and provides a molecular framework for understanding PEM.

Limitations

Our study has several limitations that should be considered when interpreting the results. First, while the SomaLogic platform offers broad proteomic coverage, it does not capture the full human proteome. At the time we contracted with the company, the 7K assay included 7288 aptamers covering 6361 unique proteins, which, although extensive, represents only a subset of all genetically encoded proteins. Consequently, pathway enrichment analyses were constrained to the proteins represented on the assay. In any event, biases are inherent to all proteomic platforms, with mass spectrometry, for example, favoring more abundant proteins rather than being limited by aptamer design.

Second, some proteins are represented by multiple aptamers targeting different regions of the same protein (e.g. extracellular vs. membrane domains). To avoid duplication in the GSEA, we selected a single SOMAmer per protein, retaining the probe with the largest absolute fold change.

Third, aptamer-based detection has inherent limitations in both specificity and quantification. SOMAmer reagents can occasionally bind protein complexes or be influenced by matrix effects. Although we relied on the manufacturer's extensive internal validation, which includes orthogonal confirmation (by an antibody-based assay, pQTL, or mass spectrometry, for example) for approximately 70% of targets (12), we did not perform orthogonal validation ourselves. The SomaLogic assay does not absolutely quantify protein levels but instead reports relative abundances.

Finally, the cohort size, while unique and valuable, was not large enough to support division into discovery and validation subsets. Replication in an independent dataset is also challenging given the intensive longitudinal design. As with all observational proteomic studies, our results represent associations rather than causal mechanisms. Future validation in larger or complementary cohorts will be essential to confirm these results and clarify the biological significance of the proteins and pathways we report.

DATA AVAILABILITY

The protein abundance data for each protein and subject and matching phenotype data is available in [Supplemental Table S1](#) and at subject, an interactive portal that provides access to research results from studies focused on ME/CFS (<https://www.mapmecfs.org>). Additional data from our analyses are available in the [Supplemental tables](#).

Supplemental data—This article contains [supplemental data](#).

Acknowledgments—We are indebted to the individuals who participated in our strenuous exercise protocol. We thank our lab manager Carl Franconi for coordinating all efforts in this project as well as the following individuals who participated in participant screening, and/or CPET, and/or blood collection: Victoria Birdsall, John Chia, Patricia Doty, Tiffany Ong, Betsy Keller, Susan Levine, Xiangling Mao, Geoffrey Moore, Maria Russell, Dikoma Shungu, Jared Stevens, Kristin Treat, and David Wang.

Funding and additional information—This research was funded by U54NS105541, an NIH grant co-funded by the National Institute of Neurological Disorders and Stroke, National Institute of Allergy and Infectious Diseases, National Institute on Drug Abuse, National Heart, Lung, and Blood Institute, National Human Genome Research Institute, and Office of the Director, by NIH U54AI178855 and by the Amar Foundation, and UL1 TR 002384 from the National Center for Advancing Translational Sciences, which provides support for the REDCap database and blood processing at the Weill Cornell Medicine Clinical and Translational Science Center. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Author contributions—M. H., L. G., M. A. E., K. A. G., and A. G. writing—review & editing; M. H., L. G., M. A. E., K. A. G., and A. G. writing—original draft; M. H. supervision; M. H. resources; M. H. project administration; M. H. funding acquisition; M. H. conceptualization; A. G. and L. G. visualization; L. G., M. A. E., K. A. G., and A. G. formal analysis; A. G. investigation; A. G. data curation.

Conflict of interest—The authors declare that they do not have any conflicts of interest with the content of this article.

Abbreviations—The abbreviations used are: CPET, cardio-pulmonary exercise test; GSEA, gene set enrichment analysis; MCS, mental component summary; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; MFI, multi-dimensional fatigue inventory; PCS, physical component summary; PEM, post-exertional malaise; TRP, Transient receptor potential; t-SNE, t-distributed stochastic neighbor embedding.

Received November 4, 2024, and in revised form, October 4, 2025
Published, MCPRO Papers in Press, November 17, 2025, <https://doi.org/10.1016/j.mcpro.2025.101467>

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